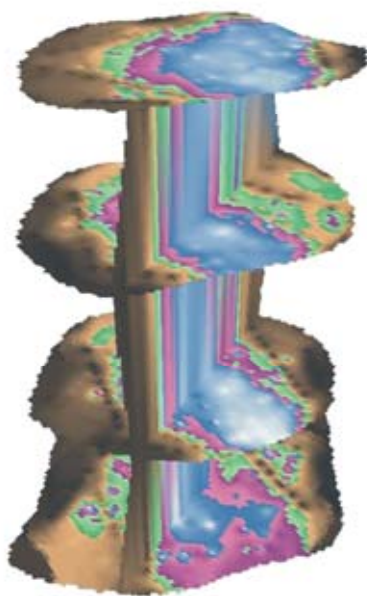


PiCUS[®]

Sonic Tomograph

Version Q72



Schalltomograph

彈性波樹木斷層画像診斷裝置

Geluidstomograaf

Tomogramma sonico

Scanner sonore

argus electronic gmbh

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PiCUS® Sonic Tomograph

Manual

Program version Q72

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1. Introduction

1.1. How sonic tomography works

The PiCUS® Sonic Tomograph is an instrument for the virtually non-invasive detection of decay and cavities in standing trees.

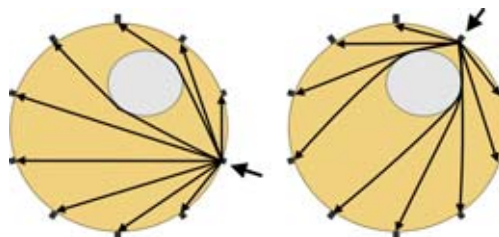
It uses the fact that the velocity of sound in wood depends on the modulus of elasticity and the density of the measured wood. Most damages, which impair the safety against fracture of trees, in particular cavities, white rots, and brown rots reduce elasticity and density in wood. These wood characteristics vary both within a tree species, and between the tree species, only large damages can be found by a comparison of individual sound velocities, which were measured on a specific tree, with tabulated standard values.

The PiCUS® Sonic Tomograph therefore uses relative sound velocities so that the system calibrates itself automatically at each measured cross-section.



The PiCUS® Sonic Tomograph consists of a set of sensors (typically 6 to 14), which are strategically placed around the tree. Each sensor is connected to a pin or nail (0.8 - 2 mm in diameter), which is tapped into contact with the wood - onto the latest tree ring - by a pin hammer at each inspection point. The sensors record the times of flight of sound waves manually induced by knocking with a small hammer. By using the times of flight of the sound wave and the distances between the sensors, apparent sound velocities are calculated.

Since every sensor records time of flight of every impact, a dense network of sound velocities across the cross-section is collected. PC or pocket PC instantly calculates a full coloured tomogram of the trees cross-section right at the tree. This tomogram gives information about the presence of decay and cavities within the tree. In this tomogram many features, e.g. remaining wall thickness and opening angles of cavities can be measured with the computer mouse.

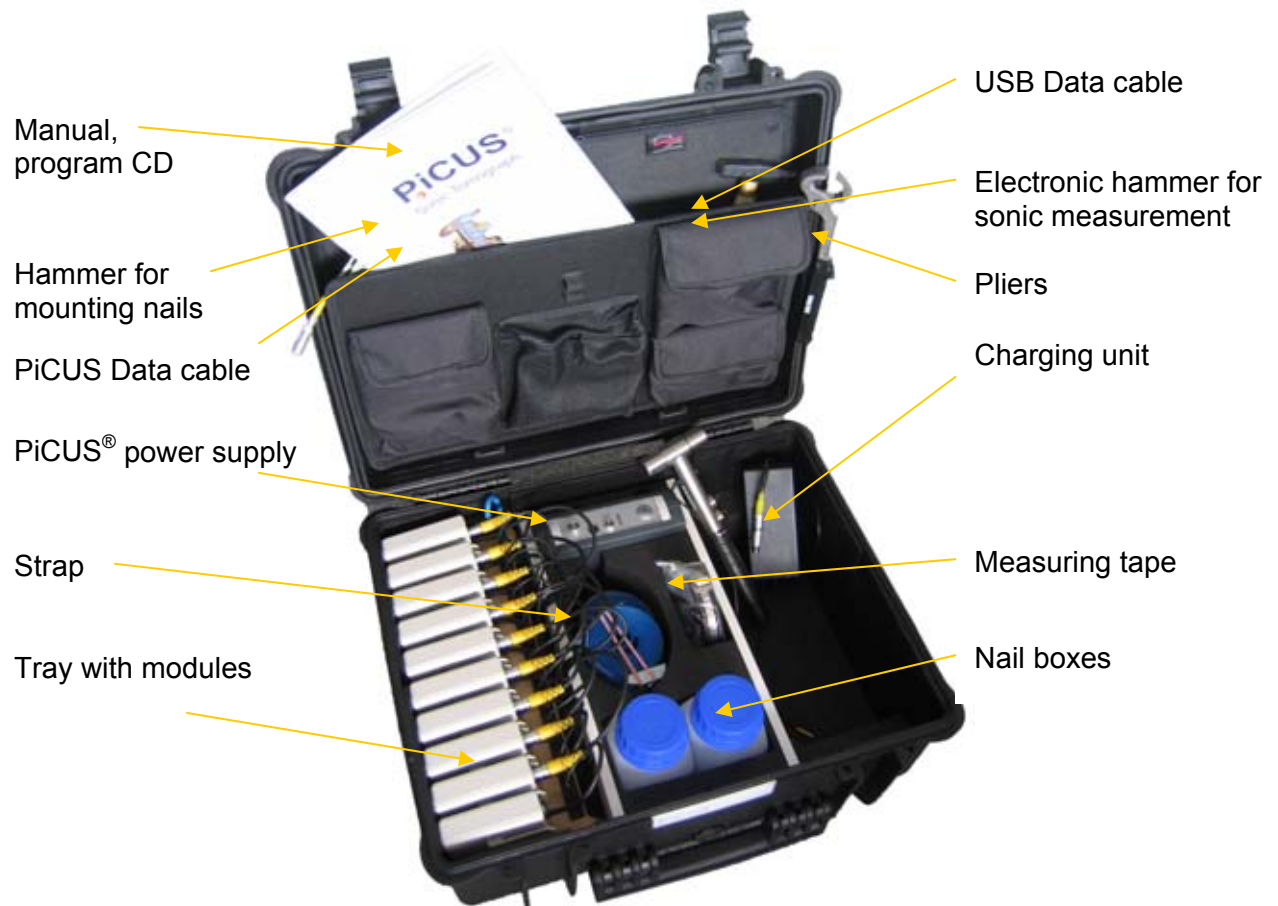


Measuring principle of PiCUS® Sonic Tomograph

2. Components of the PiCUS® Sonic Tomograph

2.1. Contents of the PiCUS® suit case

- 1 manual and CD with the P1CUS – software
- 1 hard case
- 1 measuring tape
- 1 – 2 straps with buckle
- 1 hammer for mounting nails, chrome hammer with magnet for tapping
- 1 pliers
- 2 box of roofing felt nails (used and unused nails)
- 1 PiCUS® modules (4 to 20) with cable connections and sensors on tray (number of sensors can change depending upon configuration)
- 1 PiCUS® power supply with 1 pack
- 1 cable for re-charging the batteries of the PiCUS® (black)
- 1 USB cable, connection between PiCUS and computer/laptop (black)
- 1 PiCUS® data cable, for connecting PiCUS® module chain to PiCUS® power supply



Contents of the measuring suitcase

3. Application and handling notes

3.1. Field of application

The PiCUS® Sonic Tomograph was developed for the investigation of living trees. Inappropriate application can lead to damages of the device and/or the investigated object.

Although every effort has been undertaken to present the results of measurements as generally comprehensibly as possible, they nevertheless require qualified technical interpretation.

Conclusions about the status of the examined tree are the exclusive responsibility of the user.

Neither the manufacturer nor the distributor of the PiCUS® Sonic Tomograph can be held liable for any conclusions based on measurements with the PiCUS®.

We recommend attending the training courses of the distributor or certified institutions in your country.

The PiCUS® Sonic Tomograph detects differences in the ability of the wood to transmit sound waves.

The device does not detect the reasons for the differences in the ability to transmit sound waves.

This means the instrument is not able to identify the kind of the decay – hollow, decay or cracks.

The reason for the decay is to be identified before any further work is undertaken! **Do not fell a tree because of a tomogram only. Apply the full bandwidth of arboricultural investigation methods first.**

3.2. Handling notes

PiCUS® sonic Tomograph is a precise instrument containing sensitive components.

Please handle all parts of the system with care! Pay attention to the following notes:

1. The plugs of the measurement modules are locked in the sockets. To disconnect the modules from each other please pull at the grip of the plug. Do not pull at the cable! Warranty does not cover damaged cables.



False!

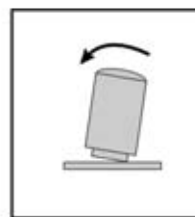


Correct

2. Sensors contain sensitive components! To generate the sonic signals you **MUST NOT knock / TAP** on the sensors. Damages, caused by improper handling of the sensors, **are NOT covered by warranty**. The maximum acceleration to apply is 4000g. This value can not be reached during normal work. Please make sure that sensors do not drop on hard surfaces (concrete or metal)!
3. Make sure that sensors do not “jump” into contact with any other magnetic material. The magnets are very strong and attract all other magnetic materials very strongly.



False!



Correct

4. PiCUS Software Installation

4.1. General information

The installation of the software consists of three main parts:

- **PiCUS PC software** is needed to run a PiCUS measurement or calculate Tomograms on the PC.
- In order to work with the PocketPC the **PiCUS Pocket PC** program must be installed on the Pocket PC.
- In order to exchange data between Pocket PC and PC a so called „**Active Sync**“ driver has to be installed on the PC.

4.2. Installation of the PiCUS® PC control and analysis software

Before you install the PiCUS program, make sure that all Windows updates are loaded on your computer. Check Microsoft's homepage for updates.

Windows XP

To install the PiCUS program you have to run the setup.exe in **Administrator login** from

Windows Vista

Start Setup.exe as Administrator by

Mouse – Right click “setup.exe”. Select “Run as administrator” in the menu.

CD:\PiCUS PC Programm\PiCUS (xx) Standard\setup.exe (to install Q71 standard) or
CD:\PiCUS PC Programm\PiCUS (xx) Expert\setup.exe (to install Q71 Expert)



setup.exe

The setup program installs

1. USB driver for communication between PiCUS power supply and PC
2. Microsoft .NET Framework
3. Direct X
4. PiCUS Q71.x

on your computer. Depending on the configuration of your computer the computer may shut down after step one and reboot. If the installation is not automatically continued please start the setup.exe again.

4.3. Registration

Each licence enables you to install the Q71x on **two** computers. Each installation needs a registration to unlock the software.

There are two ways to register the Q72x program.

1. By email (or Fax) as it used to be
2. Automatically via Internet directly in the PiCUS Program

In order to register the program through the PiCUS program the computer needs to be connected to the internet. Use an **Administrator** login for the registration.

Firewall:

If there is a firewall in your network or on your computer this firewall will block the PiCUS web service that provides the unlock code. If your firewall reports that the "picus.exe" tries to access the internet you should allow this access. Turn on "session cookies" in the "Internet Explorer" Options to use this service.

4.3.1. Registration window

After the first start of the program the following window appears:

Enter your user name here. The user name is on the sticker of the CD cover and looks like this: „PST1234-1234“

Enter your "Identnumber" here. The identification number may look like this: „fx56GW“

Enter your data in the field that are marked with this character: „*“

After pressing OK you will reach the registration window. Press "Request Authentication code". Your numbers will appear. You can now either send these numbers by email (fax) to registration@argus-electronic.de or press "Registration via Internet" button. The numbers are transferred to our web service. The unlock code is send back automatically.

The Registration window looks like this:

Push this button to create your random number code. Your code is shown in the "Key number" (2) field. These numbers are to be transferred to our service.

If the computer is connected to the internet you can use this button to use our web service (mind your firewalls). The numbers are transferred to our service and the unlock code is send back. If the computer is not connected to the internet you can send an email or Fax. Click on „Register“ to proceed.

4.4. Installation of the PiCUS PocketPC control program Q71CE

4.4.1. Windows Mobile 2005

Pre-condition for installation

Pocket version operation system: - Windows mobile 2005 or later
 - Latest version of Active Sync installed on the PC.

Connect the PocketPC to your PC by putting it into the docking station

1. Open the Active Sync Window and click on the icon to browse through the PocketPC
2. Open an explorer Window on your PC and browse to the CD directory
3. Now the PiCUS program is to be installed. Go back to the explorer window on your PC (PiCUS CD \ PocketPC Q71CE program). Double click „setup.exe“, installation will start and run automatically. Please enter the suggested paths. Please do not enter other directories.

4.4.2. Uninstall the PocketPC program

- **Deinstallation** on the Pocket PC: Start → Settings → tab „System“ → Symbol „Remove programs“
- Select „PiCUS Schalltomograph“
- Click „Remove“

4.5. Regional Settings

PocketPC and PC must run at the same regional settings. This is important to exchange PiCUS.pit files between both computers.

An overflow error can occur while opening a “*.pit” file if PocketPC and PC operate with different regional settings.

5. Sonic measurements

5.1. General Hints

Check all batteries (PiCUS radio hammer PRH, PiCUS, PiCUS Calliper, PocketPC / Laptop PC) for sufficient voltage BEFORE you leave the office!

The main steps of a PiCUS Sonic measurement are:

1. Determining measuring level and number and positions of measuring points
2. Determining geometry of the measuring level
3. Mounting the equipment on the tree
4. Sonic measurement
5. Calculation of the tomogram

This chapter is describing these procedures.

5.2. Determining the measuring level and number of measuring points

5.2.1. General notes

When starting to work on the tree the following decisions are most important:

- where to place the measuring level,
- where to place the measuring points (MP) and how many

In order to decide these questions look for external signs of internal defects like: fruit bodies of fungi, cracks, cavities, damaged bark et. Use all of your knowledge about trees and diseases. Choose the measuring level with respect to your visual assessment.

The tree needs to be investigated near ground level very often. This is because many fungi grow from the root system upwards into the trunk.

The **maximum** distance between MP: ~ 50 cm

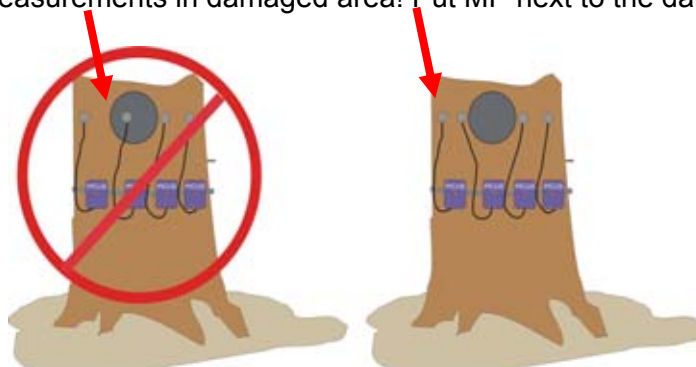
The **minimum** distance between MP: ~12 -15 cm

5.2.2. Installing nails

Use magnetic nails only! The magnet of the sensor must fit to the nail.



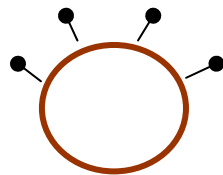
-DO NOT put MP for sonic measurements in damaged area! Put MP next to the damaged areas!



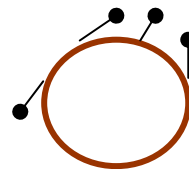
- DO put MP in damaged areas for the EIT (electric impedance measurement) measurement in order to have equidistant MP positions. This is important for EIT. Do not use those MP in decay for sonic measurements.
- The nails must touch the wood! Make sure that nails are longer than the thickness of the bark.



- All nails should point to the centre of the trunk.



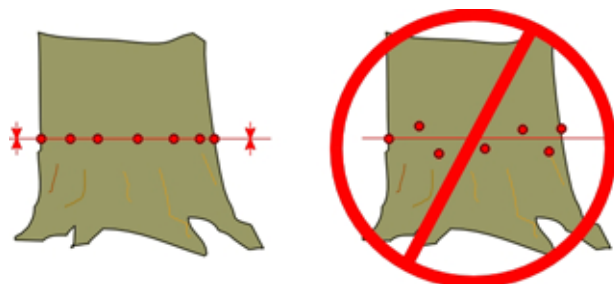
favourable



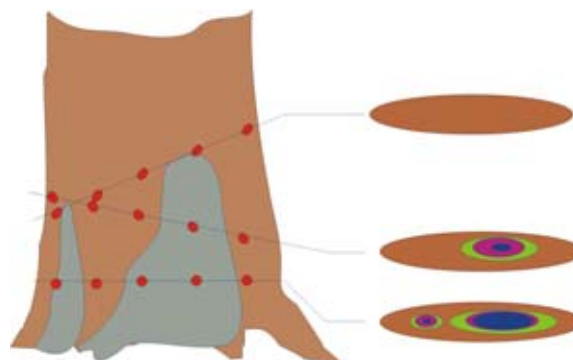
unfavourable

5.2.3. Mount MP along a line

- Tap in nails (measuring points - MP) along a (horizontal) line.



- If you intend to work with a non-horizontal line in order to get results of a larger part of the trunk mind the following cases:

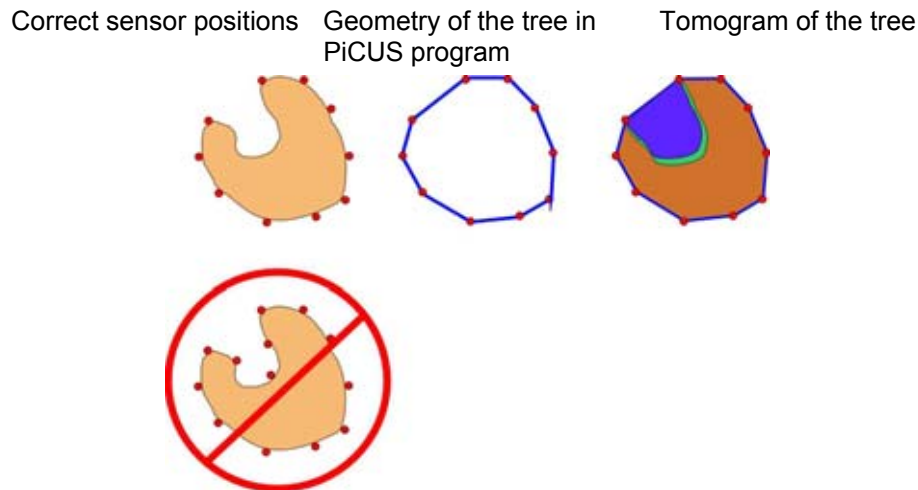


5.2.4. Installing MP on cavities and roots

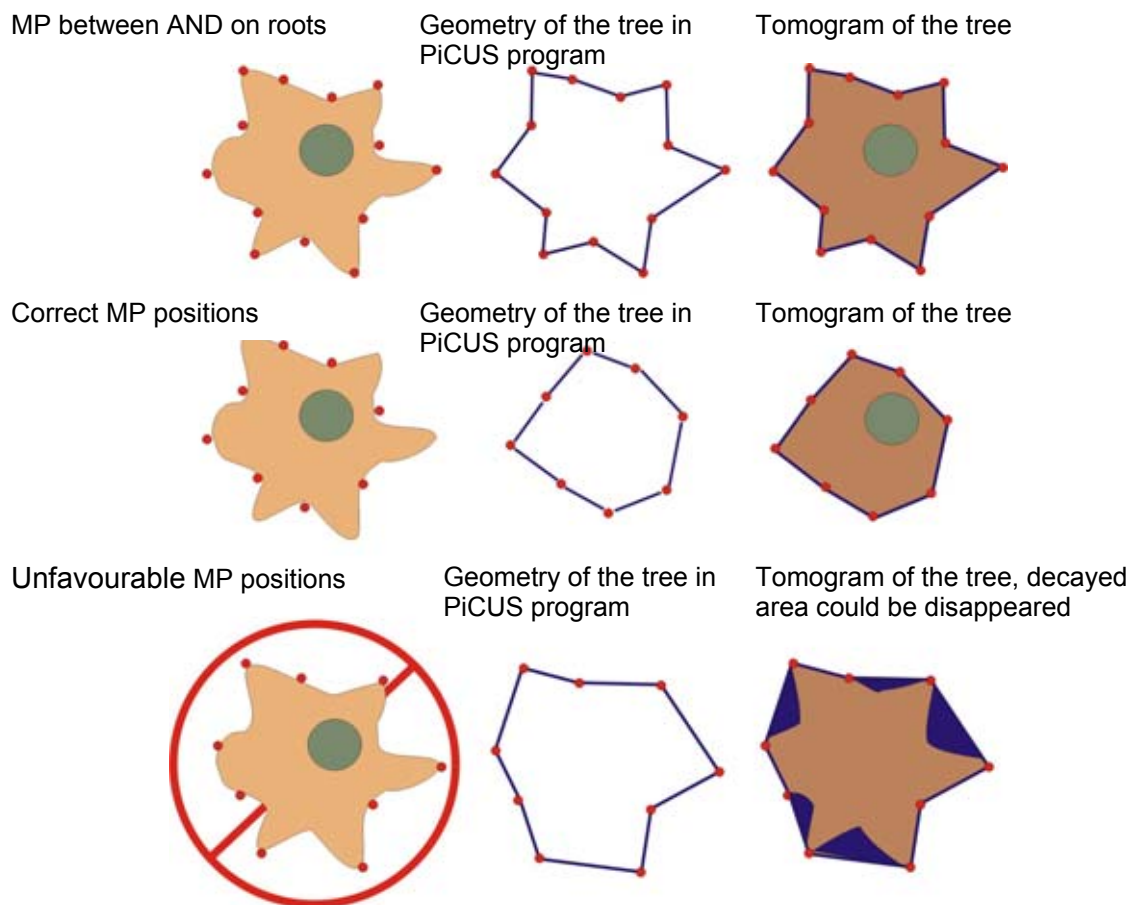
Don't install any sensors inside the cavity! Space out sensors on furthest points of the tree.

Record the geometry of the tree as usual – do not account for the cavity.

If the tomogram is calculated and the sensors are installed as shown in the sketch the cavity is shown like a „usual“ hole – indicated by violet / blue colours.



The sketch shows how to proceed on trees with larger roots.



5.2.5. Trees with two trunks

If the tree consist of two or more trunks, it is useful calculate each part of the tree separately. The reason for this is the bark between the trunks. The sonic waves can hardly travel through the bark.

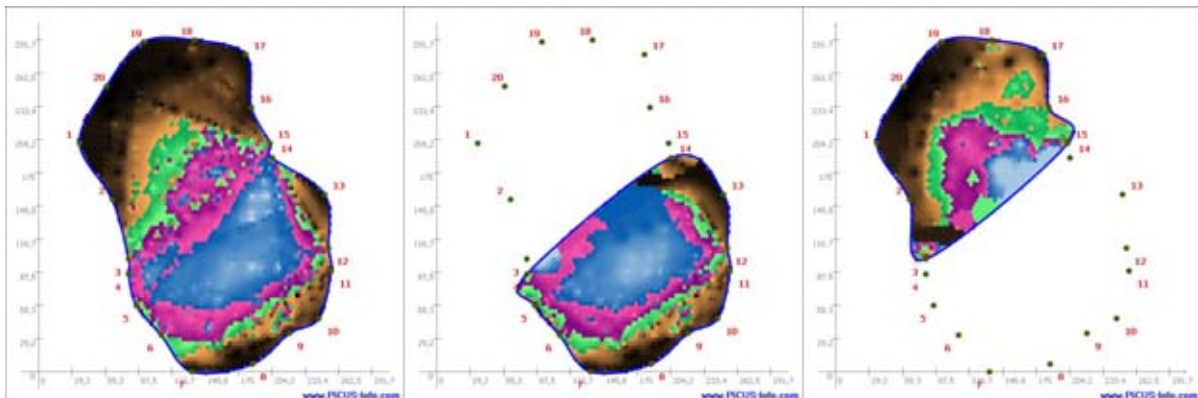
Procedure

1. Use many MP, at least 8 in every section of the tree
2. Place MP next to the location of cracks etc
3. After the sonic measurement calculate the whole tree and each section separately

Disable MP in “Receiver” and “Transmit” mode to remove it during the calculation

Histogramm	MP used	Report	F-Optionen	SIA
	MP	Transmitter	Receiver	
	1	☒	☒	
	2	☒	☒	
	3	☒	☒	
	4	☒	☒	
	5	☒	☒	
	6	☒	☒	
	7	☒	☒	
	8	☒	☒	
	9	☒	☒	
	10	☒	☒	
	11	☐	☒	
	12	☐	☐	
	13	☐	☐	

By calculating each trunk separately, the influence of the bark can be analysed. The sketch shows an example for calculating a “co-dominant stem”.



5.3. Mounting of the equipment on the tree

1. The sensors are mounted on the tree anti clockwise (top view). This means: sensor 2 needs to be placed to the right of sensor 1 and so on.
2. Wrap the strap around the tree. If you intend to measure the tree on more than one level, place the strap in a height that enables the sensor on its wire to reach more than one measuring level.
3. Mount the modules on the strap. Set the sensor on the nail. Work slowly! Do not allow the sensor to “jump” on the nail. Mind the strong magnets. Set sensor to the nail obliquely first. Remove all material from the magnets carefully without scratching on the surface.
4. Connect modules with each other. Start with module one. Plug the cable of module one into module two and so on. Protect the last connector of the chain with the cap.

5.4. Determining the geometry of the measuring level

In general the geometry of the measuring level (positions of sensors) should be recorded as accurate as possible. In a few cases, the tree geometry can be calculated by assuming an ellipse or a circle. Read chapter

“Measuring geometry of elliptical trees”.

to get information about how to do so. The majority of trees does require the more accurate “free shapes” or triangulation method. Refer to chapter

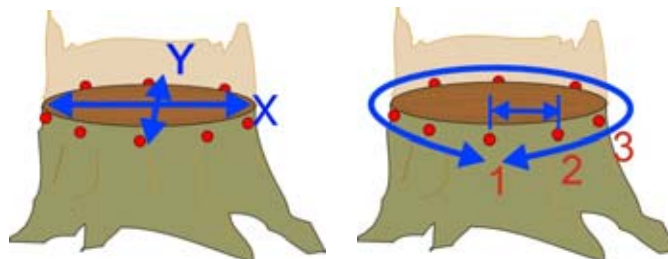
“Measuring free geometry with Calliper”.

to get information about this. In order to achieve the maximum precision, we recommend using the “free shapes” method always.

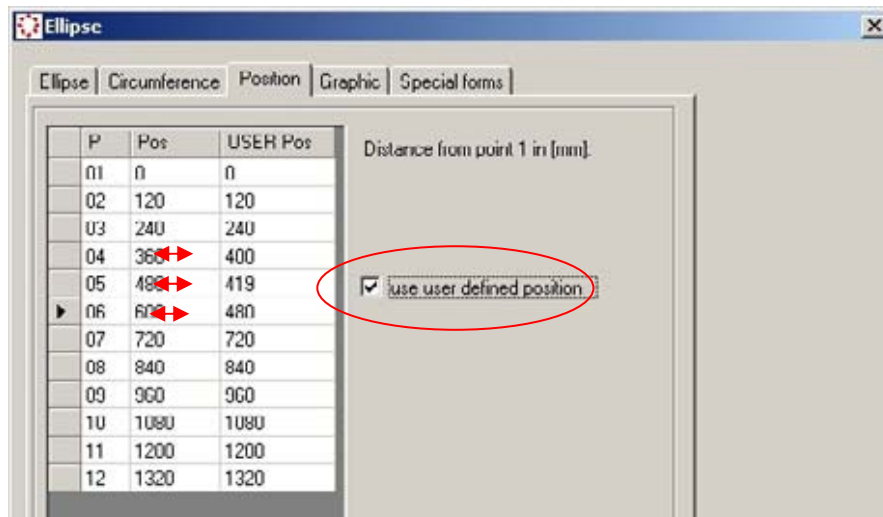
5.4.1. Measuring geometry of elliptical trees

When using this method the distance between all sensors needs to be the same.

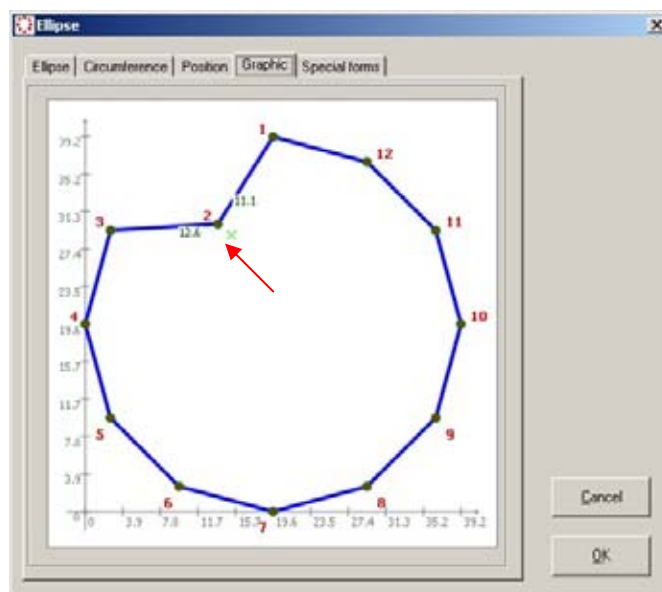
1. Determine circumference and / or the smallest and largest diameter of the trunk at the height where the measurement.



2. Fix the measuring tape around the tree at the height of measurement. The zero point (and sensor 1) coincides with the largest diameter!
3. Read off the circumference.
4. Activate PiCUS program: “start → programs → PiCUS”.
5. Create a new file by clicking: “File → New Tree Tomogram → (Type of) Tree”.
6. Open the ellipse geometry window by clicking
“Measurement → Tree geometry → Ellipse”.
7. Click on tab “Ellipse” to use the ellipse method or click on card “Circumference” to use the circumference only.
8. Select the number of measuring points (MP). The distance between MP: ~15 to 40 cm.
9. Enter the measured circumference into the textbox “measured circumference” of the PC program. Alternatively enter circumference and / or smallest and largest diameter. The larger diameter should be written into “Y” box. The smaller diameter is to be written into “X” - box.
10. The program calculates the distances between MP. You can read the positions of the nails (along the measuring tape) on the tab “Pos” (PocketPC) / “Sensor positions” (PC). If you do not want to use a constant distance between sensors, you can alter all distances in the column “USER Pos”. Set the check in “use user defined position”!



11. Click "Calculate from Ellipse / circumference" on the cards "Ellipse" / "Circumference". The computer calculates the positions of sensor along the tape.
12. On tab "Graphic" the positions of sensors can be moved by touching the screen. (mouse on PC: Left click and move). The distance to neighbour MP is shown.



13. Tap the pins into contact with the wood at the distances calculated before. (Read positions on tab "pos" - Follow the measuring tape!). **Sensors** must be mounted **anti clockwise**: Sensor 2 is right of Sensor 1 and so on.

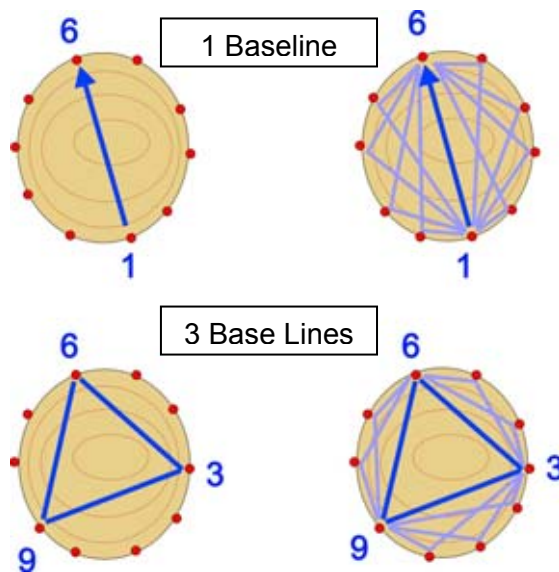
5.4.2. Measuring free geometry – triangulation method

To record the geometry of arbitrarily shaped trees the "Tri-angulations" method should be used: If we know the length of all three sites of a triangle, the shape of the measuring level can be calculated exactly.

Depending on the size of the tree and on the size of the calliper you can either use the

"One - Baseline System" or "Three - Baselines - System".

The following sketches show what distances have to be measured in both cases.



The sketch shows an example for using 10 Sensors with one Baseline and three Base lines.

The dark blue lines in the sketch indicate the baselines.

The “**PiCUS Calliper**” is an optional tool for measuring distances on the tree quickly and accurately. The photographs show the PiCUS Calliper in working and transport position.



The “Free shapes Calliper” functions can be used with or without the using the PiCUS calliper. Proceed as follows:

1. Tap sensor pins into contact with wood. Choose the location considering the situation on the tree (external signs of damages...)!
 2. Open the Free shapes Calliper window:

“Measurement → Tree geometry → Free shapes Calliper”

3. On the tab “Baseline” please enter:
 - a. Circumference of the tree (the number of MP is prompted) in [mm]
 - b. Number of MP
 - c. Number of baselines (try to work with 1 baseline first)
 - d. Orientation of the tree (e.g “Nr 1 at north position...”)
 - e. Level of measurement above ground in [cm]

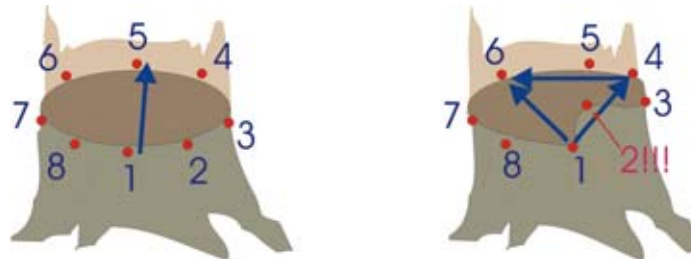
Click on “Continue”. The next tab “Distances” is shown.

4. The table on the tab “distances” shows all distances that need to be measured.
5. Select “Automatic Correction” to automatically correct triangulation errors
6. Measure all distances. Enter all distances in the table. (manually or using PiCUS calliper)
7. Click on “Calculate”. On the tab “Graphic” the tree is shown.

For all non – base points an orientation needs to be entered: is a point situated left or right of a base line.

Example: the base line runs from point 1 → to point 4. The orientation of point 2 needs to be entered. In the left sketch point 2 is situated right of the base line. That is the “standard” orientation. In the right sketch point number 2 is situated *left* of the base line.

Click on the cell in the column “R/L” to change the orientation. If a point is not in standard orientation a “!” is shown.



If the diameter of the tree is larger than the calliper try to use the “3 Baselines method”.

Sometimes it is useful to change the base points in order to reach all points with the calliper. For instance use 1, 6 and 10 instead of 1, 5 and 9.

If “Automatic Correction” is NOT selected, an error message may be shown during calculation:

“There is not triangle with these distances...”

This error occurs when the triangles geometry can not be calculated from the given figures: The addition of the two sides must be larger than the third side. Check all distances and enter the new values instead. Read Calliper manual to get additional information.

Do use the “Free shapes with PiCUS Calliper” method even if no PiCUS Calliper is available.

5.4.3. How important is the geometry?

In most cases geometry of the trunk is very important to get an accurate tomogram. In generally: the smaller the decay is the more important the geometry is.

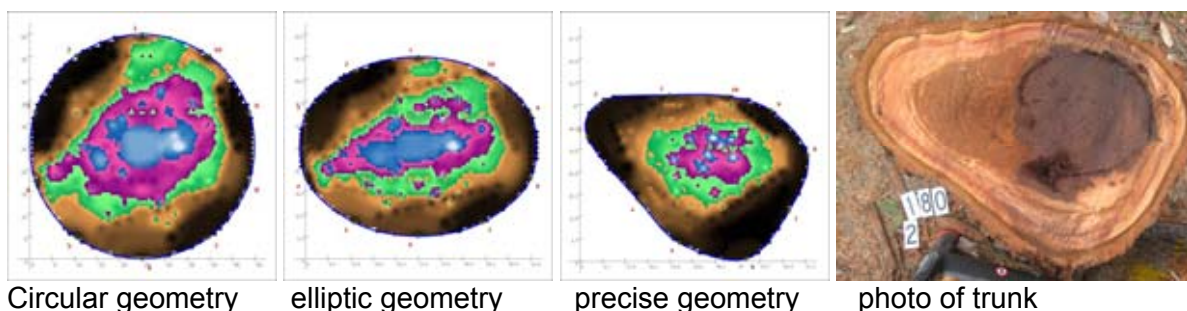
The velocity of the sound waves is calculated as follows:

$$\text{velocity} = \text{distance} / \text{time}$$

The time of flight is measured exactly by the PiCUS. The distances are calculated using the geometry data entered.

If the distance between sensors is 100cm, and the error is 10 cm, the resulting error in the velocity is already 10%. An error of 10 cm is easily obtained - just by using “circle shape” instead of “elliptic shape” or similar. An error of 10% is very much in a tomogram calculation.

The example shows different tomograms of the same sonic measurement caused by using different geometries.



Circular geometry

elliptic geometry

precise geometry

photo of trunk

5.5. Sonic measurement

5.5.1. Setup of USB - PiCUS power supply

The PiCUS power supply is connected to the PC using either wireless Bluetooth or USB-cable.

How to use the USB connection:

1. Turn OFF the PiCUS power supply
2. Install the USB cable between an USB port of a (running) PC and the PiCUS power supply
3. Turn ON the PiCUS power supply

How to use the Bluetooth connection:

1. Turn OFF the PiCUS power supply
2. Remove USB cable from the PiCUS power supply
3. Turn ON the PiCUS power supply

The PiCUS power supply shuts down automatically after 20 minutes of idle time.

5.5.2. Sonic measurement with chrome hammer and Y cable

Please find version Q71x of the manual on the PiCUS CD to read about using the Y cable.

5.5.3. Sonic Measurements with electronic hammer (PEH / PRH)

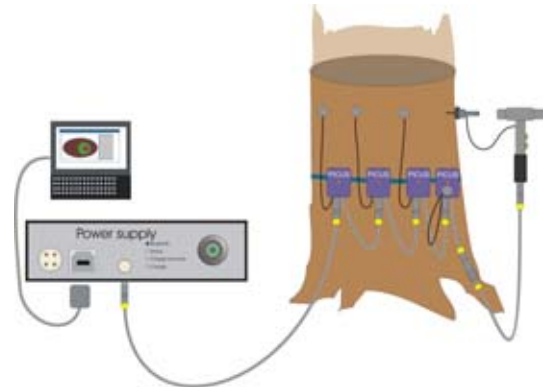
The standard way of the sonic measurements requires the use of one sonic sensor on each measuring point (MP). A measuring point is a nail.

The use of the electronic hammer (PiCUS Electronic cable Hammer (PEH) and PiCUS Radio Hammer (PRH)) offers to use fewer sensors than measuring points (MP).

PEH and PRH can also be used if number of sensors is equal to the number of MP.

5.5.4. Hardware setup for PiCUS electronic hammer – PEH

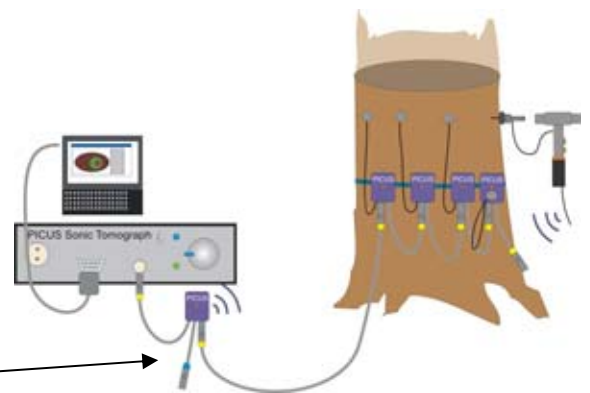
Connect the data cable (**7 pin socket female – 7 pin connector male**) to the electronic hammer (PEH) and to the last module of the module chain.



5.5.5. Hardware setup for PiCUS Radio Hammer (PRH) and old power supply

PiCUS power supplies build before 7/2007 can also be used with the radio hammer PRH.

An additional receiver – module will be supplied. This module needs to be included into the module chain, either at the beginning or at the end. Read chapter “PiCUS Radio Hammer PRH – additional notes” also.

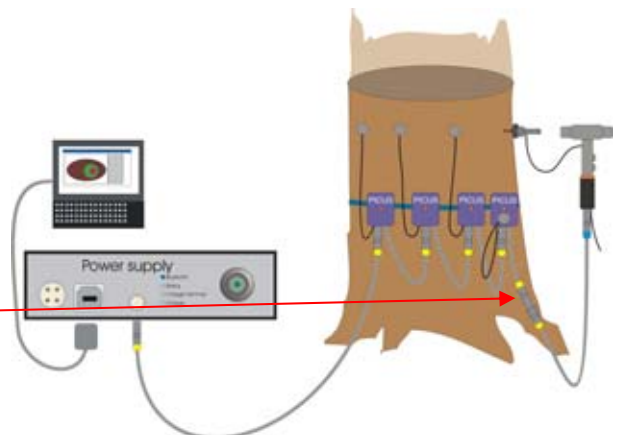


5.5.6. Hardware setup and operation for PiCUS Radio Hammer – PRH

The PiCUS radio Hammer (PRH) can be operated wireless (using a 422 MHz technology) or cable based.

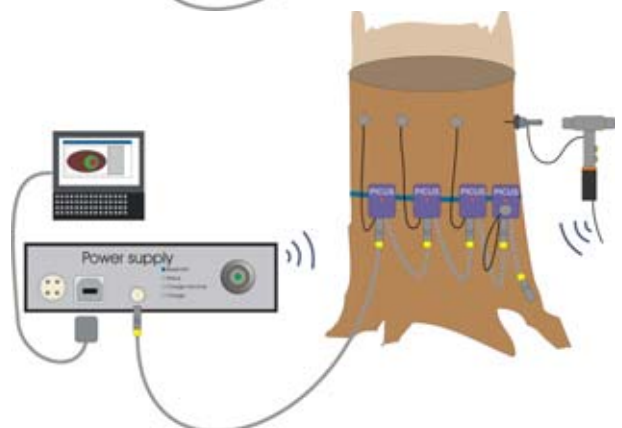
How to use the cable connection:

1. Wait until PRH is turned OFF (automatic shutdown after 7 seconds)
2. Install the special hammer cable (**7 pin female – 5 pin male**) between handhold of PRH and last module of the chain
3. Press either button on the PRH



How to use the wireless connection:

1. Wait until PRH is turned OFF (automatic shutdown after 7 seconds)
2. Remove the cable from the PRH
3. Press either button on the PRH
4. Amber LED on power supply is on: good radio connection Hammer - PC



The hammer turns itself off after approximately 10 seconds of idle time. The amber status LED of the power supply is turned off.

5.5.7. PiCUS Radio Hammer PRH – additional notes

There is a 422 MHz Radio connection between the PRH and the PiCUS power supply. This wireless communication may be interfered by:

- other devices using the same frequency (thermometer etc)
- strong electromagnetic field (for instance close to electric railways of light rail (tram way) etc)
- large trees in-between the hammer and the PiCUS power supply

An indication of a weak radio connection is the status LED on the power supply (second from top).

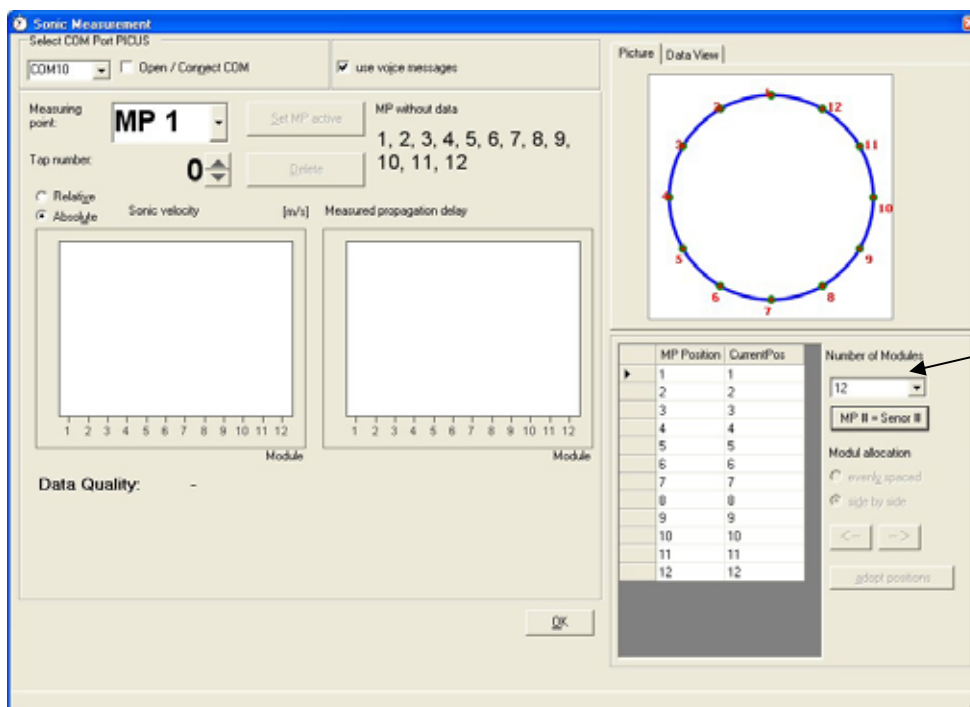
- It shows solid light while radio connection to the hammer PRH is active (hammer is ON)
- It flashes one time when either button on the hammer is pushed.
- It flashes continuously when the connection between PC and power supply is interrupted.
- It flashes in-constantly when the radio connection to the hammer is weak.

The “weak connection problem” can be solved by moving the power supply (the PiCUS case) closer to the tapping position. If this does not help, the hammer PRH needs to be connected to the end of the module-chain by the special hammer cable: **7 pin male – 5 pin male**. The **5-pin connector (blue) fits into the hammer handheld**.

It may also be necessary to use the cable connection to PRH if the distance between tapping point and receiving points is smaller than approx. 15 cm.

5.5.8. Sonic measurement with electronic hammer PEH / PRH

1. Connect the Pocket PC / PC with the PiCUS power supply by using the USB cable if you want to use the USB connection. For the wireless Bluetooth connection the USB cable must NOT be connected.
2. Start the sonic measurement “Measurement” → “Measure Sonic”. Enter a file name when you are prompted to do so. The standard PiCUS data directory is “my documents\PIt”

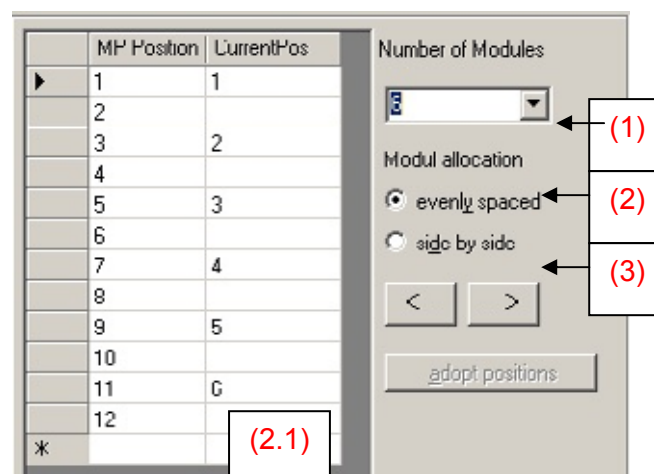


3. Select the number of sensor modules that you want to work with, typically 9 to 12. The number of sensors needs to be equal to the number of MP. You may either select the number from the drop down list (1) or use the button “MP# = Sensor #”. If you intend to use fewer sensors than MP please refer to the next chapter “Fewer sensors than measuring points”.
4. Turn the PiCUS power supply on.
5. Open the serial port by setting the check next to the COM selection box “COM open / connect”. Turn “sound on” at your computer to hear the voice that guides through the measurement.
6. Press the buttons on the hammer to specify the “active” measuring point. Make sure that the active MP is the one that you intend to tap on. Wrong allocation will void all data! Tap gently on the first measuring point (nail) at least 3 times.
7. Move hammer (and pin) to the next measuring point (nail). Press the buttons on the hammer to specify the “active” measuring point. The voice from the computer will read the current position when the pin was tapped or when buttons were pushed. Make sure that the active MP is the one that you intend to tap on. Wrong allocation will void all data!
8. Tap on ALL measuring points (MP)!

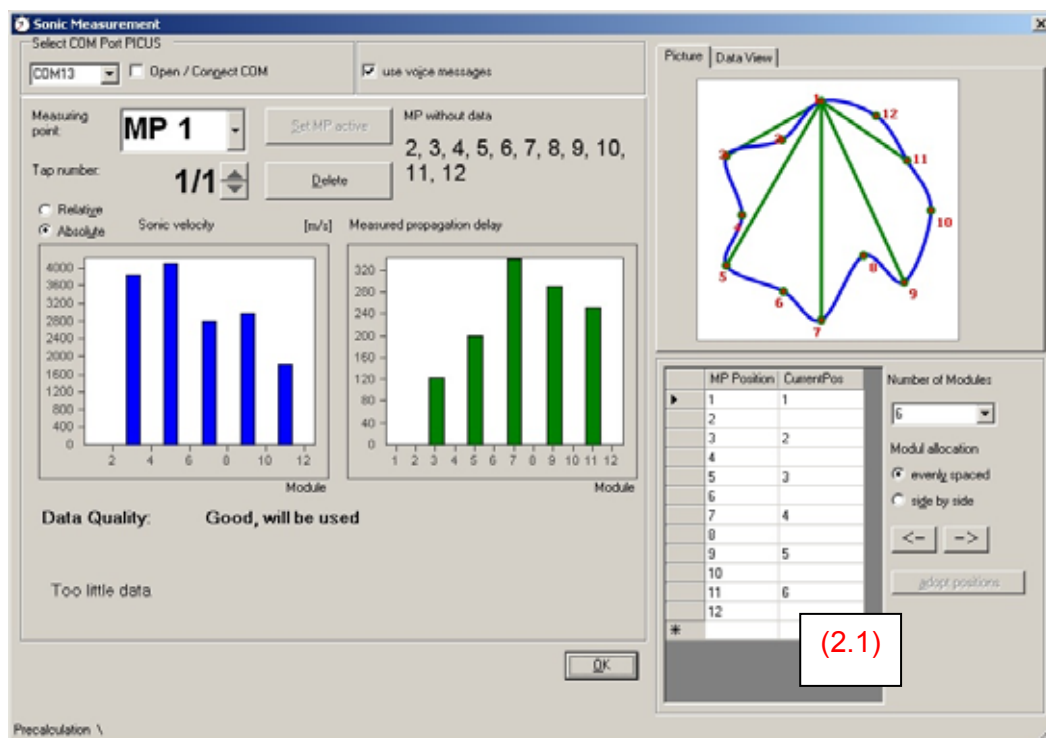
5.5.9. Using fewer sonic sensors than measuring points (MP)

The use of the fewer-sensors-than-measuring points – technology requires the use of an electronic hammer (either PEH or PRH).

1. Connect the Pocket PC / PC with the PiCUS power supply by using the USB cable if you want to use the USB connection. For the wireless Bluetooth connection the USB cable must NOT be connected.
2. Start the sonic measurement “Measurement” → “Measure Sonic”. Enter a file name when you are prompted to do so. The standard PiCUS data directory is “my documents\PIt”
3. Select the number of sensor modules that you want to work with. This example shows the procedure for 6 sensors and 12 measuring points. Select the number (in this case: 6) in the drop down list (1).
4. Specify the location of the sensors on the tree by using the “evenly spaced” (2) , “side by side” (3) button and/or the table (2.1). You can edit the table with mouse and keyboard. Press button “adapt positions”



5. Place sensors on MP/nails like specified in the table (avoid mistakes!!!):
 - Measuring point (MP, Nail 1) – Sensor Nr 1
 - MP 2 – clear
 - MP 3 – Sensor Nr 2
 - MP 4 – clear
 - MP 5 – Sensor Nr 3
 - MP 6 – clear
 - MP 7 – Sensor Nr 4 etc
6. Turn the PiCUS power supply on.
7. Open the serial port by setting the check next to the COM selection box
“COM open / connect”. Turn “sound on” at your computer to hear the voice that guides through the measurement.
8. Push either button on the hammer one time to activate the hammer (LED is flashing quickly). Press either button on the hammer to specify the “active” measuring point. Make sure that the active MP is the one that you intend to tap on. **Wrong allocation will void all data!** Tap gently on the first measuring point (nail). Screen should look like this. Only MP's with sensors attached will give readings.



Tap at least 4-5 times.

9. Move hammer (and pin) to the next measuring point (nail). Press either button on the hammer to specify the “active” measuring point. The voice of the computer will read the current position when the pin was taped or when buttons were pushed. Make sure that the active MP is the one that you intend to tap on. Check screen every time before you tap! Wrong allocation will void all data!
9. Tap on ALL measuring points (MP). Tap on nails with AND without sensor! Check position every time the tapping point is changed.

This procedure is called “first round of taping”. You may run the calculation after this step. In many cases the tomogram will look close to the final result. To collect all data completely you have to proceed as follows:

10. Move all sensors to the next MP:

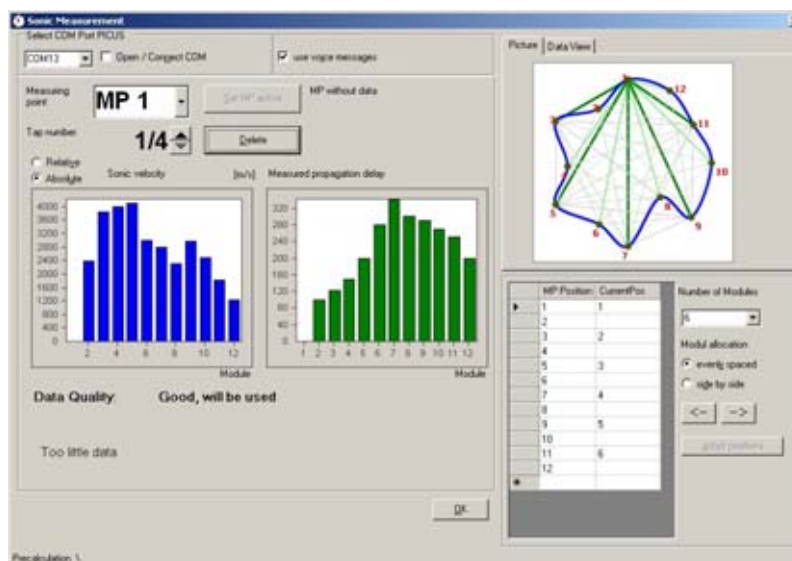
- MP 1 – clear
- MP 2 – Sensor Nr 1
- MP 3 – clear
- MP 4 – Sensor Nr 2
- MP 5 – clear, Etc

11. Specify the new location of the sensors in the table. You may either use the “Arrow” buttons next to the table or enter the positions by hand (2.1). The table should look like this:

	MP Position	CurrentPos
1		
2		1
3		
4		2
5		
6		3
7		
8		4
9		
10		5
11		
12		6
*		

Number of Modules:
 Modul allocation:
☒ evenly spaced
☐ side by side

12. Move hammer (and pin) to MP 1 (nail 1). Press either button on the hammer to specify the “active” measuring point. Tap on MP1, at least 4-5 times. The screen should look like this. (You need to select tap nr 1 to see the complete set of data.)



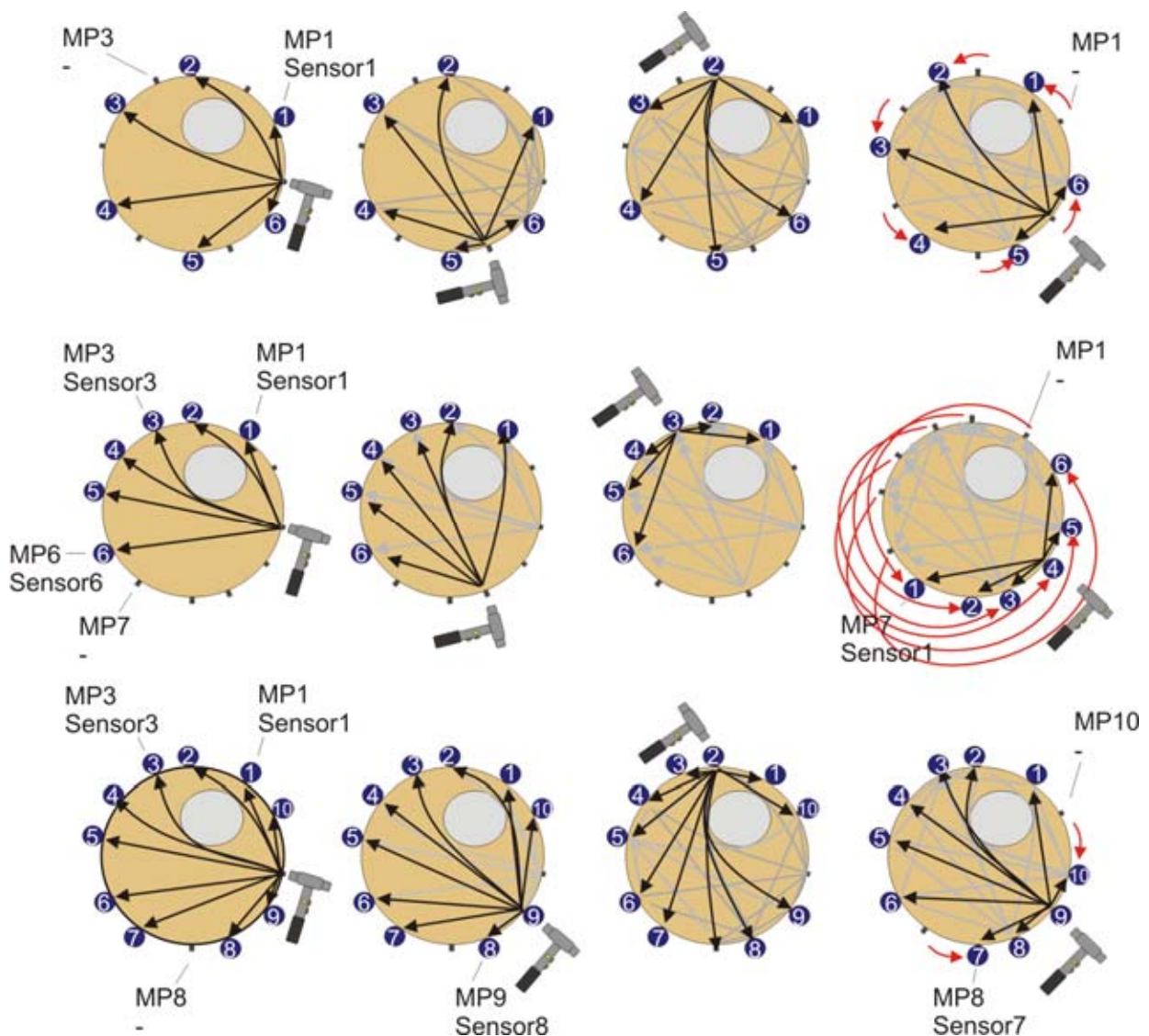
13. Tap on ALL measuring points (MP). Tap on nails with AND without sensor! Do not forget to press buttons on the hammer each time you move to another nail.

14. Click OK to close the window.

5.5.10. Fewer sensors than MP - Additional notes

- If you are using just one or two additional MP, the tomogram can be calculated before the sensors are moved to the next positions. The result will be acceptable in many cases. For example: 14 MP, 12 Sensors. For example, MP 2 and 8 do not have sensors. Tap on all 14 MP. The result will be good - very likely without moving sensors to MP 2 and 8.
- If only some sensors need to be moved to the "new" MP, the other sensor can stay on their „old“ positions. They can stay in the module chain.
- Try to use always as many sensors as possible.
- Despite of the Fewer – Sensors – than – measuring –points technology, do not use too many MP. Too many MP cause too small distances between MP! The distance between MP should not under-run 12 to 15 cm. Otherwise there is the risk of generating calculation errors because of errors in geometry measurement.

The sketch gives three examples of how to place sensors.



5.5.11. Technical notes of PEH / PRH

- The PEH hammer can only be used for tapping as long as the LED is flashing quickly!
- The LED flashes slowly some seconds after a button was pushed and after the pin was tapped. The hammer PEH is in-active when LED is flashing slowly.
- The hammer PRH is shutdown when the LED is not showing light. To re-activate the hammer press any button. The position will not be changed if the button is pushed while the LED is flashing slowly (off).
- **Mind the cable on the hammer (data cable or antenna). The cable must not touch another sensor or sensor cable. This will interfere data acquisition!**

5.6. Calculation of the Tomogram

In order to calculate click “Calculation” → “Calculation B” or the symbol



You can find more information about the calculation in chapters “Software description” and “Tomogram calculation”

5.7. After the Measurement

1. Terminate PiCUS - program on your PocketPC: “file” → “exit”.
2. Switch off the PiCUS.
3. Disconnect the long data cable from PiCUS and module 1.
4. Collect modules, beginning with the last (e.g. number 10 or 12), and mount on tray. Put modules directly back on the tray.
5. Pull out pins/nails with pliers. Disinfect nails before using it again.

5.8. Spreading of an infection by using dirty nails

There is the possibility that a tree disease spreads from tree to tree by using the same nails. For this reason, DO NOT USE nails more than once without disinfection.

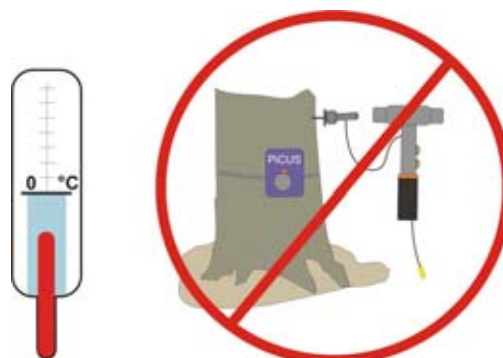
Disinfection methods: Heat ($\geq 130^\circ\text{C}$, at least 15 min), High percentage alcohol (or disinfection liquid) for many hours.

5.9. Environmental conditions conditions for sonic measurements

The components of the PiCUS® Sonic Tomograph are not water resistant. Please do not use the system in hard rain. You may use it in very light rain. Shut the measuring case when doing so. Do not cut cables!

The properties of the wood of a tree depend on many factors. One of these factors is the temperature. If the temperature is below 0°C ($32^\circ\text{Fahrenheit}$, 273°Kelvin) for a longer period of time the properties of the wood change dramatically.

Therefore, the PiCUS® Sonic Tomograph is not to be used on trees during or after periods of frost.



6. PiCUS PC Software description

6.1. Introduction

The PiCUS Software contains functions to proceeding measurements, calculating results and presenting results of:

- PiCUS Sonic measurements on a tree, to get information about the trunk or large branches
- PiCUS Sonic – Roots measurements to get information about the location of the roots of a tree, about a single root or about the root – soil system.

6.2. PiCUS Data directory

The PiCUS data files are saved by standard in the following directory:

C:\My documents\PIt.

Other locations will work also.

6.3. Menues

6.3.1. Menu item “File load”

By using the „File“ → “Load” command a dialog is opened that shows all PiCUS files. Valid PiCUS files have the extensions: „.pit“, “.trt“, “.pir“ or „.b3d“.

„.pit“ - „PiCUS (Sonic) Tomograph“. Files of Sonic tomography measurements.

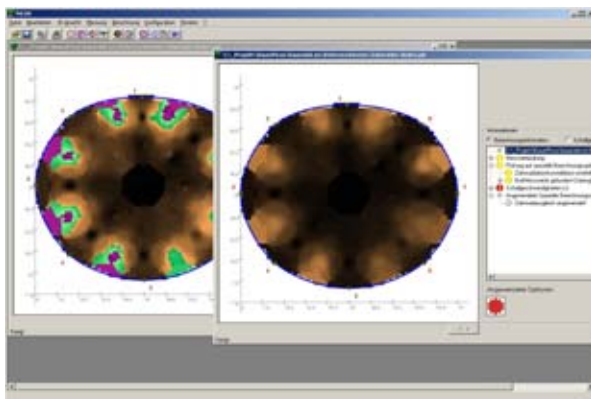
“.trt“ - “TreeTronic”. Files of electric impedance measurements.

„.pir“ means: „PiCUS Root (Finder)“. Files of this type are produced when using the PiCUS hardware for finding large roots below the surface near the tree. More information can be found in the chapter „PiCUS Root Finder“.

„.b3d“ means: „Baum (tree) 3 Dimensional“ presentation file.

You can load more than one file (.pit, .trt, or .pir data files, photos etc) at the same time into the PiCUS program (Expert versions only).

The **calculation options for each window** can be set **separately**! The same data file can be opened in several windows! This means: different calculation options can be applied to the same file in different windows. The influence of the calculation options can be analysed this way. When opening a file the default calculation options are used. The use of faulty options is avoided this way.



This example shows the PiCUS sonic file of a linden tree with a typical cog wheel data constellation (background). In foreground the same file is shown with cog wheel compensation applied.

You can also load photographs or tomograms into the main window.

6.3.2. Menu item “Save Geometry data only”

This file save dialog is used to save only the geometry data of a file. This function can be used if the tree is to be investigated in several levels with nearly the same geometry.

6.3.3. Menu item “New 3D view of sonic Tomograms”

This function is used to present all tomograms that has been recorded on the same tree in a three dimensional graphic. This function is first of all supplied for presentation purposes. It helps to give untrained persons an idea about the internal structure of the tree. The colour between the measuring levels is interpolated.

Example

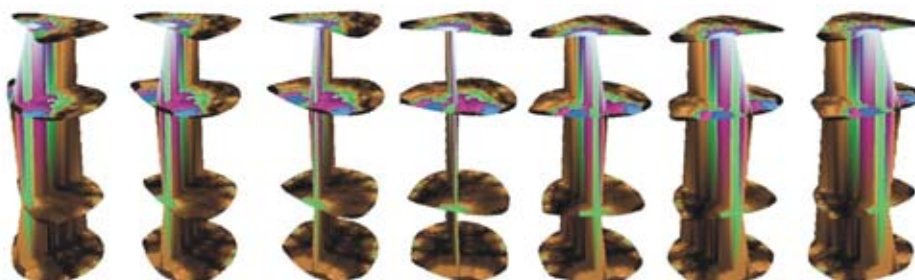
The following eucalypt tree was investigated in four levels. The investigation showed clearly that there is a connection between both infections and therefore that the trunk is separated in two parts.



In order to calculate a 3D graphic proceed as follows:

1. Measure the tree in at least two levels. Distance between levels: 25 to 50 cm. Place MP-numbers on top of each other: example: MP 1 of the 30 cm level is precisely above MP 1 of the 10 cm level.
2. Click “File → New 3D View of sonic tomograms“. The 3D window is shown.
3. Click on “Add file” to add at least two files of a tree to the 3D project.
4. Select a file in the list and click on configuration to view/alter calculation parameters for each file separately if necessary.
5. Click “Calculate” to run the calculation for all files loaded.
6. After calculation you can save the tomograms in a 3D file (”.b3d“) by clicking “File → Save as”.
7. Click “3D” to open a new window and to see the twisting tree. Twisting options are shown by clicking F1. When pressing “p” the current graphic is copied into the right part of the 3D window.

By “right click” on the right graphic in the 3D window you can toggle between the “cut image” and the 3D graphic.



6.3.4. Menu item “Edit → Tree data”

In the form “Tree data” you can write down notes about the tree like:

- Species (botanic names)
- Measuring height
- Location of the tree
- Tree Expert etc.

The list of botanical tree names is located in the PiCUS program directory.

C:\program files\argus electronic\PiCUSQxx\Bin\ PicusTreeData.txt“

You can alter or extend the list. You should use a simple text editor like “notepad.exe”. Do not use a word processing program.

Stick to the following file format:

„Scientific name / German Name / English name / French name / Netherland name / Italian name / Spanish name / typical velocity Minimum / typical velocity Maximum“

Use separator character „/“ even you do not enter the names in other languages:

Example:

```
# Scientific / DE / ENG /F/NL/I/ESP/vmin/vmax
Quercus robur / Stiel-Eiche / English Oak / / / //
```

6.3.5. Menu item “Configuration → Settings“

Some options of the calculation can be altered in the settings window. If this window is called when a PiCUS data file is opened, the changes done take effect on the opened file in the active window only.

If no PiCUS data file is opened the changes do effect all files opened afterwards. Most changed are volatile when the program is terminated.

When opening the same file in more than one window the influence of the options of the calculation can be analysed.

The list shows some of the options that can be altered in the settings window:

- Standard settings for the user
- Configuration of the legend
- Resolution of the tomograms
- Activation of sound and voice messages

6.3.6. Menu item “Configuration → Select COM port”

Select the serial COM ports for PiCUS Sonic Tomograph and PiCUS Calliper in this window. Some systems allow using different COM ports for both devices.

6.3.7. PiCUS command icons



File open

Save file

Copy the tree graphic into the Windows clipboard

Print Tomogram

Enter (or Change) geometry of circular or elliptic trees

Enter (or Change) geometry of irregular shaped tree (Free shapes, old version)

Enter (or Change) geometry of irregular shaped tree using the PiCUS Calliper (Free shapes)

Sonic velocity measurement

Calculation of the tomogram, Method B

Calculation of the tomogram, Method C, bee-lines between measuring points

Draw a radius for remaining wall thickness (1/3 – red circle)

Show / hide minimal remaining wall thickness according to Wessolly (green circle)

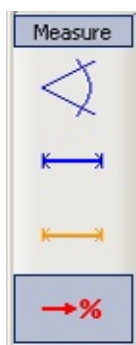
Show / Hide the legend

Show / Hide wind rose

Show / Hide prevailing wind indicator

Show / Hide description icons

6.3.8. Measuring Icons



Measure angles in the tomogram (left click)

Measures distances

Measures distances (different colour)

Rough estimate of the remaining strength of the cross section at this point using the geometrical moment of inertia. Use left mouse button and click at the green/brown border in the tomogram to show this value.

Double click deletes all measurements.

6.3.9. Diagnostics Icons

Diagnostic icons can be used to enter the visual observations into the "pit" file.

Load icon into the tomogram: left click

Move icon in the tomogram: right click and choose "move" option

Diagnostics	
	Open Cavity
	Crack
	Co-dominant stem
	Bark inclusion
	Decay
	Fruiting body of fungus (Type 1 – Toadstool Fungus)
	Fruiting body of fungus (Type 2 – Bracket fungus)
	Effluent
	Callus, strong growing wood
	Bark anomaly
	Mechanical (bark) damage (for instance because of car impact)
	Tension load
	Compression load
	Compression load
	Damaged root
	Foreign material (steel, concrete etc)

7. Tomogram calculation and Interpretation

7.1. General hints

Although every effort has been undertaken to present the results of measurements as generally comprehensibly as possible, they nevertheless require qualified technical interpretation.

Conclusions about the status of the examined tree are the exclusive responsibility of the user.

Neither the manufacturer nor the distributor of the PiCUS® Sonic Tomograph can be held liable for any conclusions based on measurements with the PiCUS®.

We strongly recommend attending the training courses of the distributor or certified institutions in your country.

The PiCUS® Sonic Tomograph detects differences in the ability of the wood to transmit sound waves.

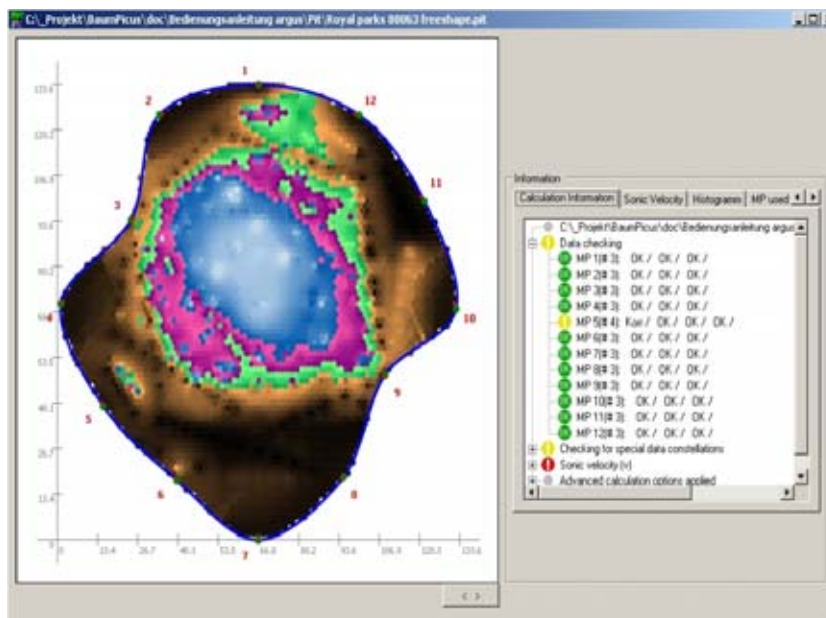
The device does not detect the reasons for the differences in the ability to transmit sound waves. This means the instrument is not able to identify the type of the damage: it may be: either hollow, decay or cracks. The reason for the decay is to be identified before any further work is undertaken!

Do not fell a tree because of a tomogram only. Apply the full bandwidth of arboricultural methods first.

The tomogram can be calculated on a PC or Pocket PC. The calculation on a PC is faster and more accurate.

Hint: because of the limited resources of the Pocket PC not all calculation functions are applied in the Pocket PC program. That is why the tomogram calculated on the Pocket PC might vary a little from the tomogram calculated on the PC.

After the calculation is finished information about data checking and the calculation itself can be reviewed on the tab cards next to the tomogram.



7.2. Tab-page “Calculation Information”

The tomogram calculation consists of four main parts:

1. Data checking
2. Pre - processing of data (velocity calculation, normalizing velocities etc)
3. Detection of special data constellation
4. Calculation and drawing of the tomogram

The event-tree is showing the results.

7.2.1. Data checking

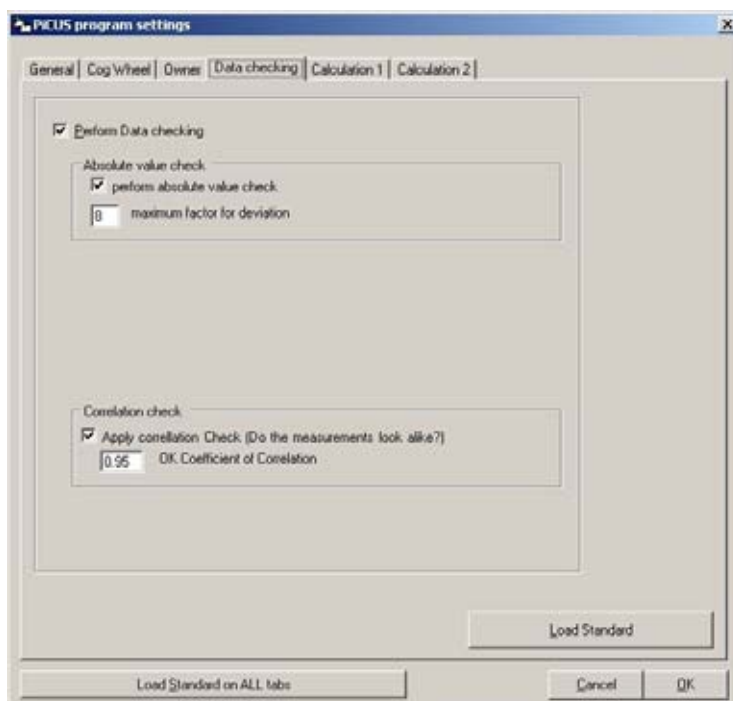
The sonic data collected is tested for errors automatically. Most of the errors are found. The following functions are used while data checking:

- Absolute value test (Are the runtimes measured more than x times larger than the average?)
- Correlation test (Do the measurements look alike?)

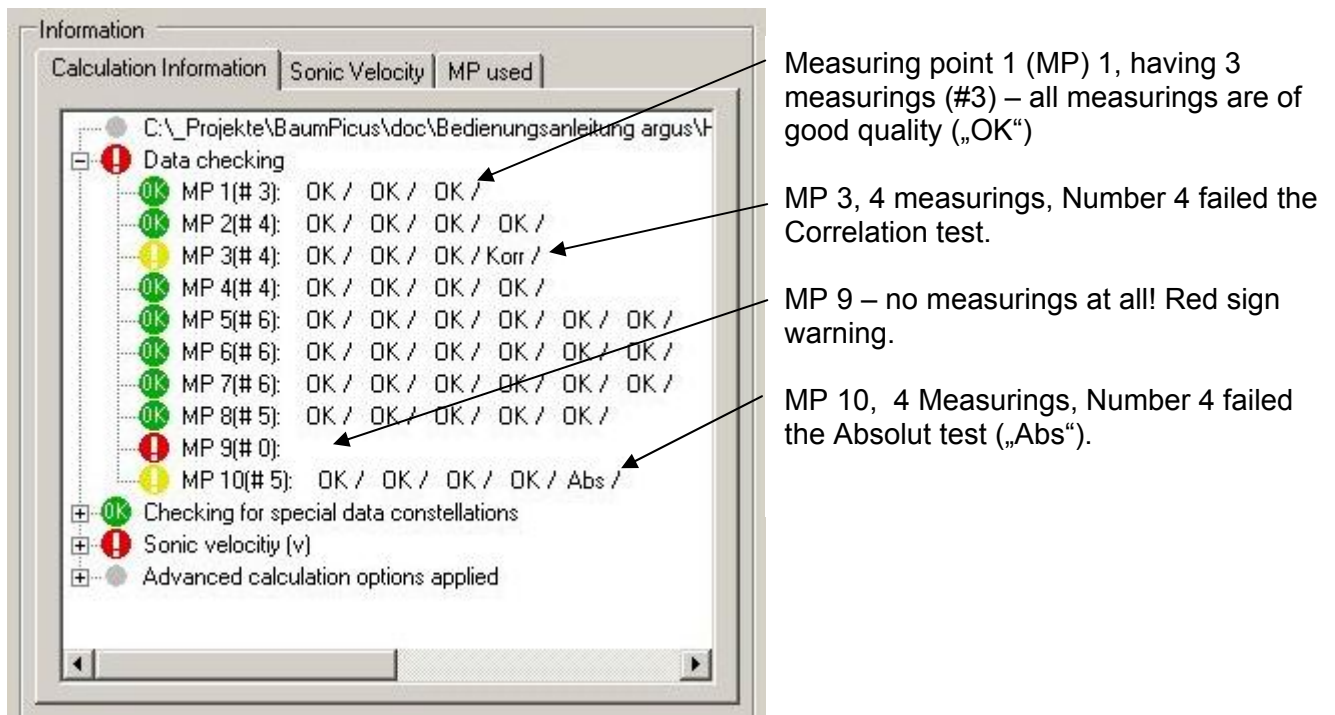
Program version Q7x Expert enables the user to activate and de-activate these tests in the configuration window:

“Configuration → Settings → Data Checking”

All settings of this window take affect in the active PiCUS “.pit” window only! A “.pit” file can be opened in more than one window. For each of these windows different settings can be used.



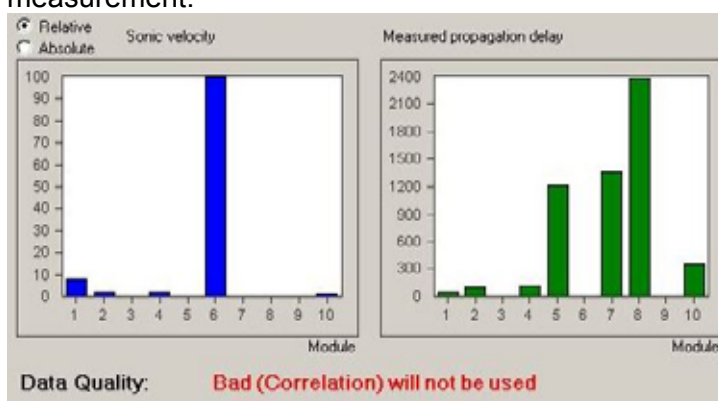
The results of the datacheck are shown here.



If the tomogram does not show an understandable result, you need to check the sonic data “by hand”. In order to analyse a file “by hand” proceed as follow:

- Backup the original file first
- Open the file in the PiCUS program
- Open the measuring window (hammer symbol)
- Check all tapings on every measuring point

The left bar graph shows the relative or absolute sonic velocity of the measuring [meter per second]. The right bar graph shows the time of flight [approximately μSec] of a faulty measurement.



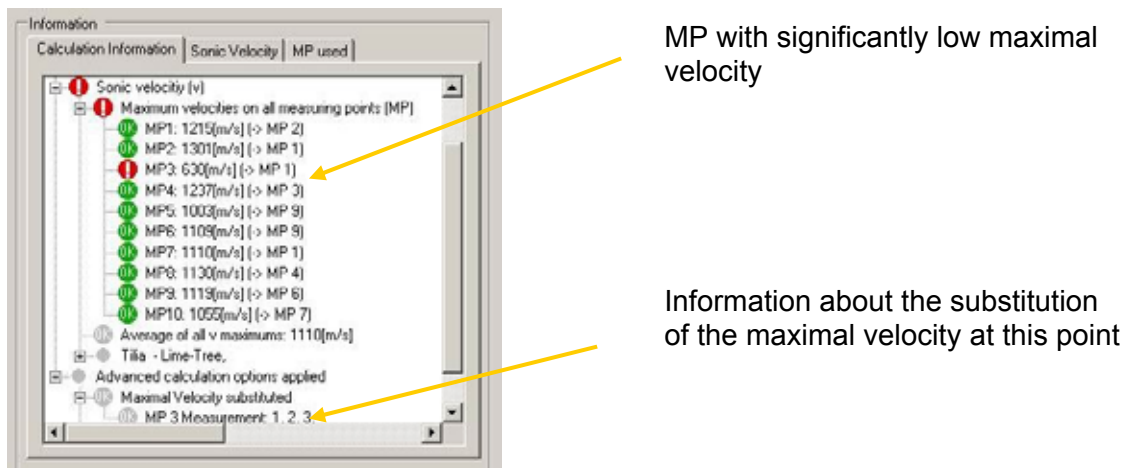
7.2.2. Maximum sonic velocity supervision

The “maximal sonic velocity supervision” test compares the maximal sonic velocity occurred on each measuring point with the average sonic velocities of all points. There are two main reasons for the velocity to be very low (or high) :

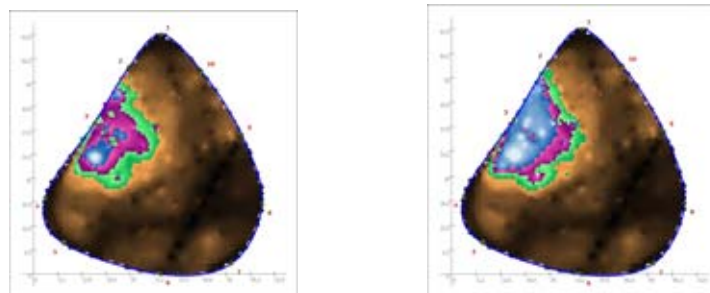
- the nail is in bad contact to the wood (put it in deeper)
- the sensor was placed in or closed to a damaged area.

If you are still on site you should make sure that all nails were placed correctly. If this was done and the calculation still reports this problem a compensation is applied as follows. Instead of using the fastest local velocity, the average of all fastest velocities is used for the 100% reference at this point.

The result of the check and the velocity “substitution” is shown in the information part of the window:



The following example shows a Linden tree several years after car impact damage, left tomogram without and right tomogram with maximum velocity supervision.



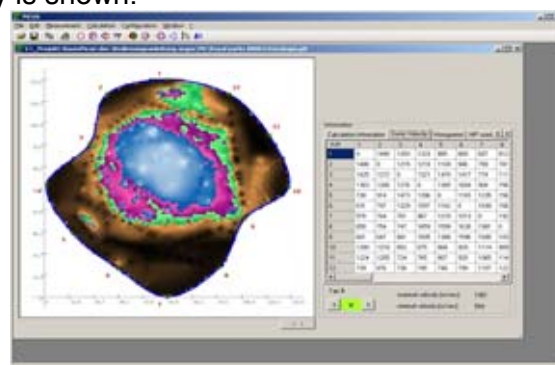
Sensor number 3 was placed in damaged wood.

Examples for this test can be found on CD:\..\PiT\Velocity supervision.

- "Linde172 Anfahrsschaden.pit"
- "Kiefer 010cm.pit"
- "Christ's Hospital 48.pit"

7.3. Tab-page "Absolute sonic velocity"

The tab-page "Sonic velocity" shows the absolute time of flight of every measurement and the velocity that is used for the calculation (the average). When the "v" (velocity) background colour is green, the average velocity is shown.



Please mind that sonic velocities depend on many conditions. Some important are:

- Species of tree
- The measuring instrument
- Measuring „line“ in relation to the shape of the tree (along the circumference or straight through)
- Season, water contents, nail length etc

Therefore the values recorded will differ from values in literature.

7.3.1. How to export the velocity table:

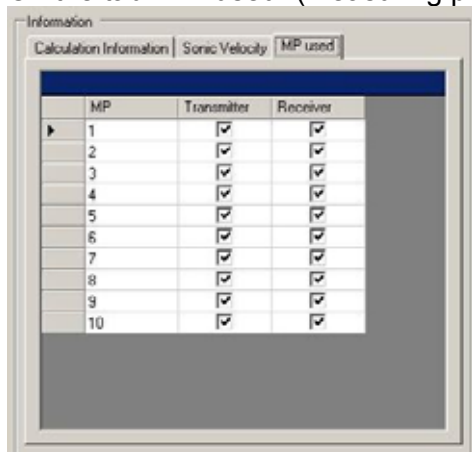
Right-click on the table and select a function from menu.

The data of the table can be copied to the windows clipboard to past it directly into other applications or to a text file (extension *.txt).

7.4. Tab-page “MP used”

In rare cases it might be necessary to exclude single sonic data from the calculation. A typical case is when a sensor was in bad contact to the wood because the nail was not deep enough or the nail was placed in a decayed area.

On the tab “MP used” (Measuring points in use) the use of MP can be specified.



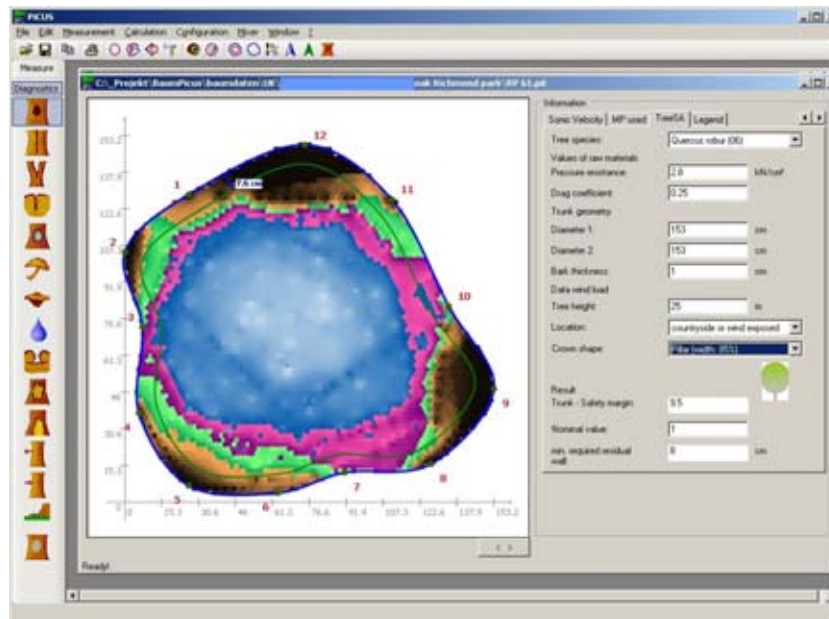
Remove the check in column „Transmitter“ to ignore all data that was recorded while tapping at this MP.

Remove the check in the column „Receiver“ to ignore all sonic data received on this MP.

Remove both checks to deactivate the MP completely. Neither sonic data nor geometry data will be used.

7.5. Tab-page “TreeSA”

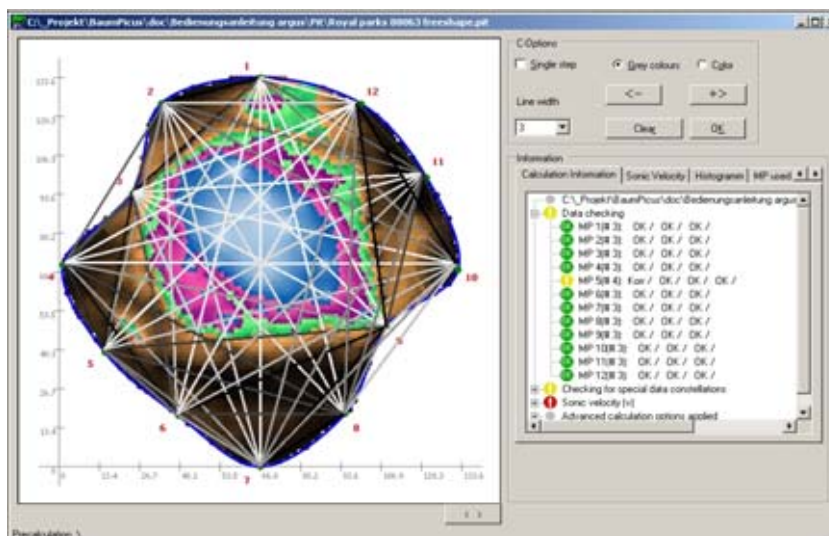
“TreeSA” means Tree Stability Assessment. The TreeSA calculations are based on the publications of Dr Wessolly, Günther Sinn and Martin Erb, published in “Handbuch der Baumstatik” *). TreeSA incorporates tree size, tree species, material properties of green wood of this species, environment conditions and crown shape to calculate the minimum remaining wall thickness.



7.6. Calculation C

“Calculation C” shows the bee-lines of all sonic data. The lines are colour coded: The darker the colour of a line is the faster the sonic waves travelled.

If calculation C is called after the calculation B the tomogram is overlapped with the bee-lines. This image is useful to identify artefacts.



7.7. Calculation B – new test version

The PC program version contains a new (test-) version of calculation B. Calculation B (version 2000) is used by standard.

The new test version “CalcB2007” can be activated by

Configuration → Settings → tabulator „Calculation 2“ → Frame „Version“

„Max v difference warning level“ should be set to 10%.

It is recommended to use both calculations.



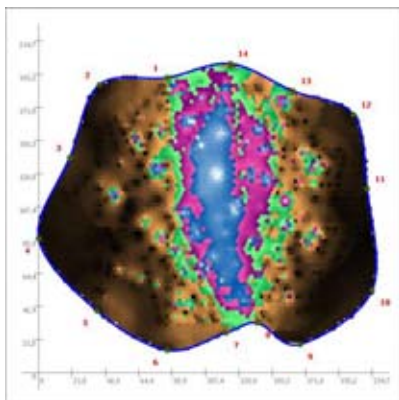
Calculation „CalcB 2007“ should be used for

- Cracks, in particular for end-to-end cracks
- Two-trunk trees
- Off –centre damages

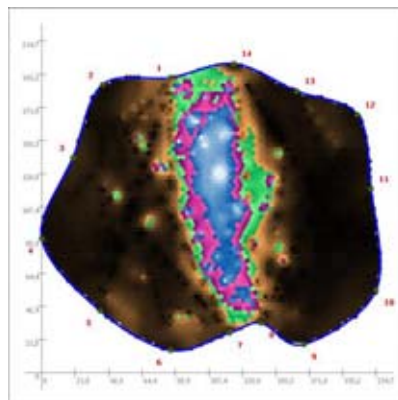
Calculation „CalcB 2007“ should NOT be used on trees

- With early stages of decay
- Decay that does not clearly change wood properties (Ustulina)

The following example shows the improvement of the tomogram on a beech with a crack.

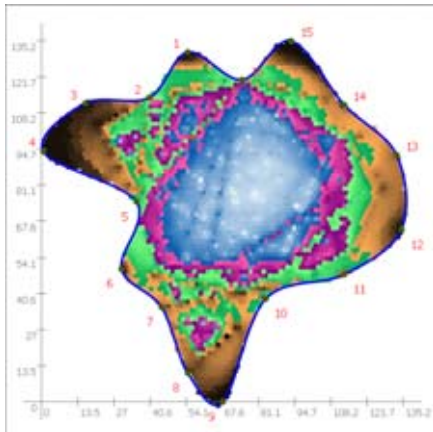


CalcB 2000

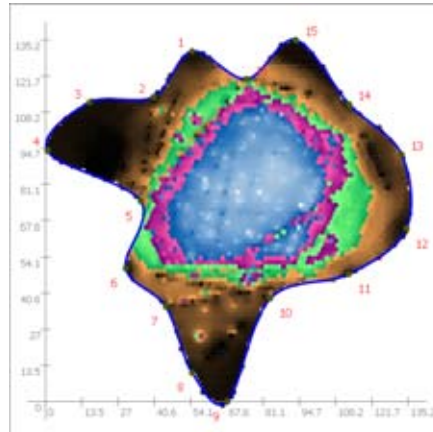


CalcB 2007

CalcB2007 shows the quality of roots better on tree if roots have been recorded accurately:



CalcB 2000 – wrong tomogram



CalcB 2007 – better tomogram

7.8. Detection of special data constellations

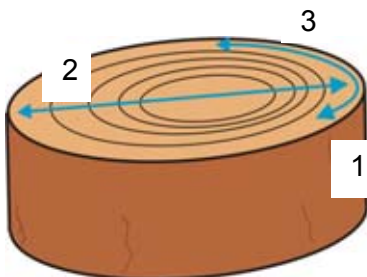
The properties of wood in living trees sometimes have sonic data that cause the tomogram to look different to that expected.

Some of the effects can be detected and compensated by the PiCUS program.

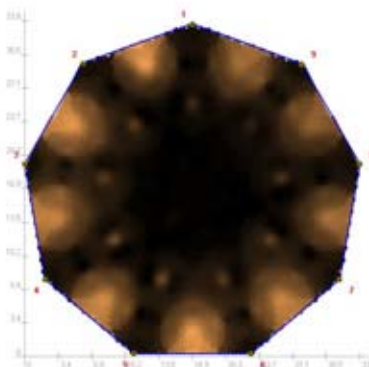
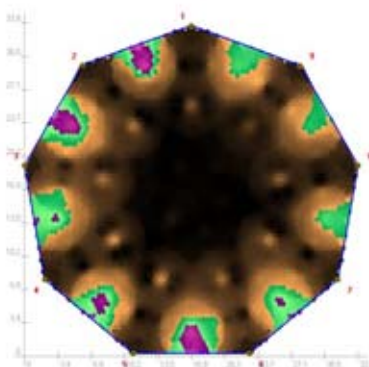
7.8.1. The “cogwheel” effect

The reason for what we call “cog wheel” effect are the different sonic wave transmission properties of the inner and outer areas of the trunk.

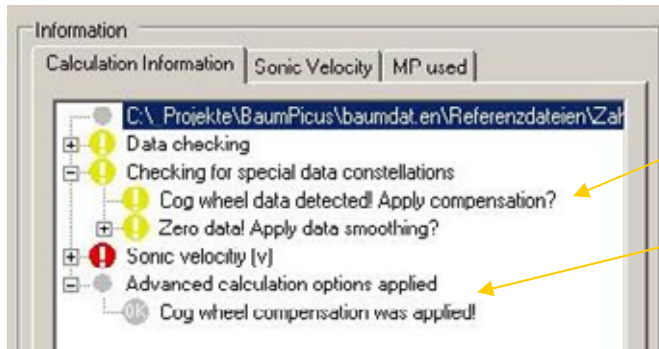
The arrows in the sketch show three different ways the sonic waves do travel. The sonic signals travel best in vertical direction (1) – along the trunk. The second best transmission direction is radial (2) - straight through the trunk. The slowest direction is tangential (3) – along the circumference.



Because of these properties the tomogram of a sound tree sometimes looks like a “cog wheel”. The green / violet area between the sensors indicates a lower sonic velocity along the circumference. The example shows a conifer tree (pine): left tomogram shows “cogwheel”, the photo on the right shows the pine tree stump.



Due to our experience we know that the wood between sensors is not damaged in cases like this. If the PC program identifies the cogwheel - data constellation the colour scale of the tomogram is shifted to hide this effect. The tomogram in the middle shows the result with cogwheel compensation applied.



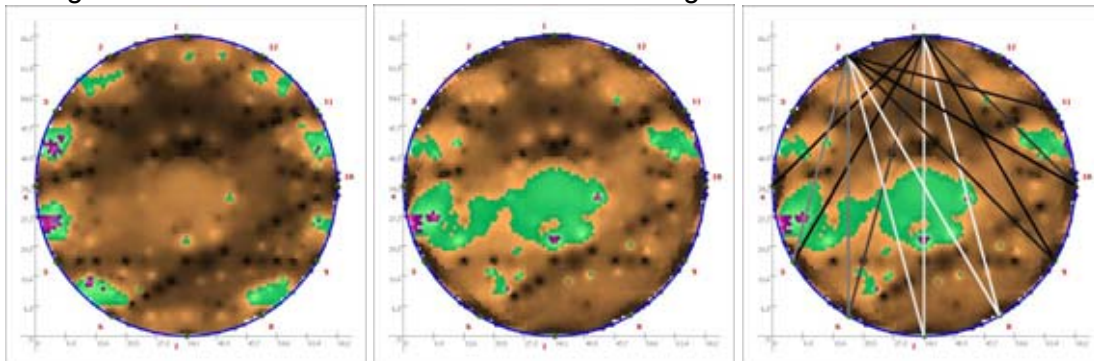
Cogwheel data constellation detected

Compensation applied because of settings in the configuration window.

7.8.2. Exclude the neighbours of the tapping point

The cogwheel data constellation can hide early / smaller defects in the centre of the tree in rare cases. In situations like this the neighbours of the tapping point – the receiving sensors that cause the cogwheel – can be excluded from the calculation.

The left tomogram was calculated with standard settings. The tomogram shows a cog wheel but the light brown centre of the tree is not normal for cog wheel data constellation in sound trees.



The central tomogram was calculated without two neighbours on both sides of the tapping point. The right tomogram is showing the lines that were used for calculation on two points: Line 1 - 2, 1 - 3, 1 - 12 and 1 - 11 were not used!

The option to exclude neighbours from calculation can be found on

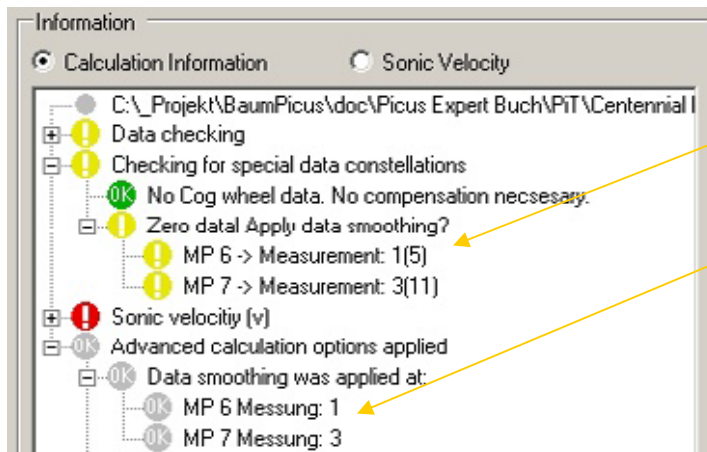
“Configuration → Settings → Calculation 1”

The Example was recorded by Dirk Pellens and David Claes from “www.interplant.be” (Belgium). It shows a populus tree with COSSUS infection.

7.8.3. Smoothing option – Zero readings

On large damaged trees the data quality of the sonic measurement may be lower than usual. This means that it is difficult to get three similar readings or some sensors do not receive sonic signals at all. In some of this cases the use of the “smoothing” option may be useful. The measuring data is “calmed down” a little by using this option: the fast readings will be slowed down a little and the slow reading will be a little faster. The PiCUS program checks the data file for zero readings during the calculation. If zero readings are detected the use of the smoothing option is suggested. The smoothing option is to be used **in rare cases only**.

You should check the suspect readings in the sonic measurement window before using this option.



Zero readings detected on MP 6, at measuring # 1 there are 5 receiving MP with zero readings

Because of the settings in the configuration window the smoothing option was applied to the following measurements: MP 6, tap # 1 and MP 7 tap # 3.

The behaviour of the smoothing function can be altered in the configuration window:

„Configuration → Settings → Calculation 1 → Data Smoothing“

You can choose if you want to apply the smoothing function to:

1. all readings,
2. to readings that have a zero value on ONE receiving MP
3. to readings that have zero values on TWO or more receiving MP

In many cases it is useful to use the second option. But in general the it is difficult to predict the best option.

Examples for the use of the smoothing options can be found on CD:

CD:\..\PiT\Smoothing option

The smoothing option is disabled by standard because **the use** is recommended **in a very few cases** only!

7.8.4. Specify minimal colour space

The colour space of the PiCUS sonic tomograms is stretched between the slowest (light blue colour) and fastest (dark brown colour) velocities recorded. In rare cases both slowest and fastest velocity are nearly similar. This causes a disturbed tomogram, showing more details than relevant for the interpretation.

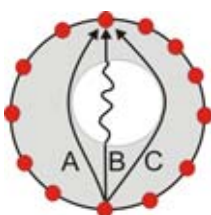
To avoid this, the minimal size of the colour space is 20%.

To de- activate the minimal size of the colour space remove the check at:

“Configuration → Settings → Calculation 1 → Set minimal colour space”

7.9. Reading the tomogram

The PiCUS® Sonic Tomograph detects and shows differences in the ability of the wood to transmit sonic waves. The ability of the wood to transmit sonic signals does strongly correlate with its density. On the other hand the sonic velocity does not necessarily vary in a measuring level: depending on the type of defect (decay vs. crack etc) the sonic waves might either been slowed down in a lower density area (early decay) or they have to bypass a cavity at (nearly) constant velocity. In both cases it is difficult to decide which wave reached the receiver first. For this reason it should be considered to speak about the *ability of the wood to*

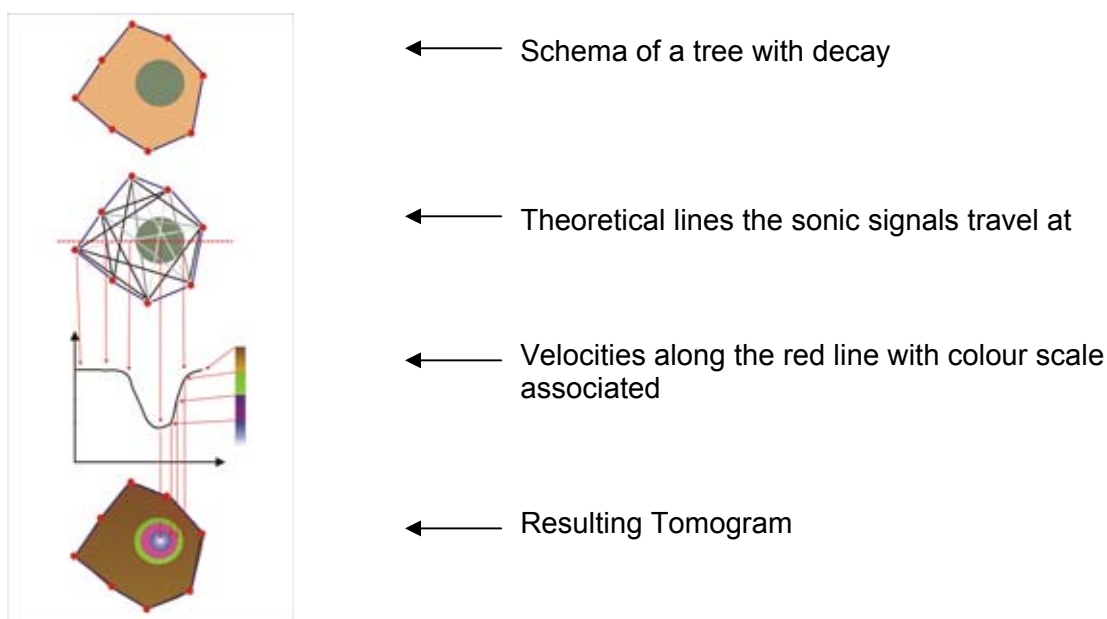


transmit sonic waves instead of *velocity* or *density maps* when analysing tomograms.

In order to read the tomograms correctly a **sound knowledge of trees, tree diseases** as well as about the **working principles of the PiCUS** is needed.

7.9.1. Colours of the tomogram

In order to show the different qualities of the wood the PiCUS program uses artificial colours. After calculating relative velocities one colour is associated with each point of the tomography level. Areas of high density (where the fastest velocities can be found) are represented by black (dark brown) colour. White / blue colours represent areas of low-density i.e. slow velocities. The colour scale (black, brown, green, violet, blue, white) is expanded between the 100% velocity and the slowest velocity. There are special trigger levels for switching from one colour to another. The following sketch (not to scale) shows the general procedure.



There are three main colours to be taken into consideration:

1. Black / brown
2. Green
3. Violet /blue/white

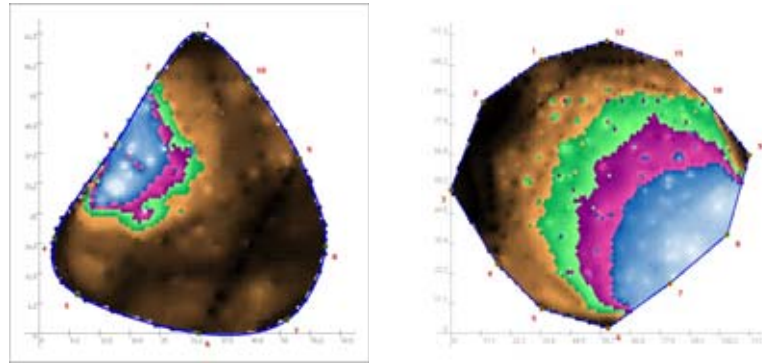
The colours violet blue and white need to be treated as one class only. For this reason the drawing of the blue and white colours can be suppressed on demand.

In general the result of sonic based measurement can not give information about the type of the structural loss in the tree directly. This means the tomogram does not tell us if a cavity, or a crack or a decay is causing a violet area in a tomogram.

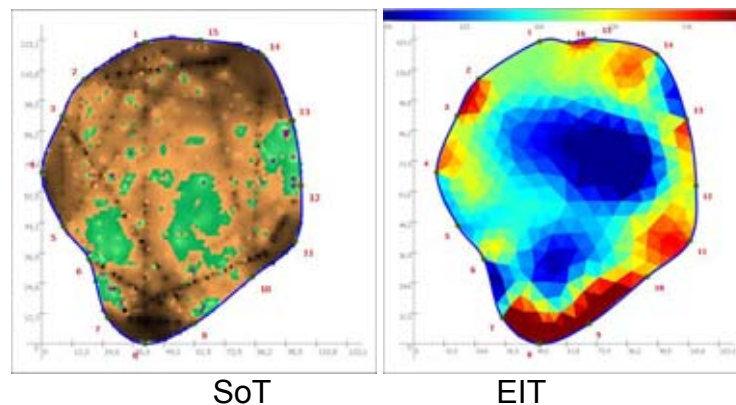
Nevertheless we can obtain additional information about the type of the damage by observing how the colours are spread (colour allotment) across the tomogram. In all cases a sound knowledge of trees and tree – diseases is needed to interpret the tomograms correctly. Experiences in operating the PiCUS and the experiences of other tomograms are very useful also.

Examples:

1. We can assume that the reaction zone (barrier zone) around a damaged area of the tree are well built if the main tomogram colours (brown, green, violet) lay very closed to each other. The left tomogram shows a typical case: the very small green area between the violet and brown area indicates a sudden increase of density.



2. In contrast to that, the right tomogram shows a wide green area and a relative long distance between the brown and the violet area. The tomogram was recorded on a tree with a Ganoderma infection. Having a tomogram like this the tree does most likely not have strong reaction walls. The fungus did probably break reaction walls frequently and is still very aggressive.
3. The following tomograms (SoT left, EIT - Elektrisches Impedance Tomogram - right) of a beech tree are typical for an infection with kretschmaria deusta.



The blue area in the EIT shows clearly the fungus infection. A sound beech has higher resistivity (red colours) in the centre of the trunk.

The SoT shows areas of violet and green dots in a mostly brown cross section. By watching this pattern we can say clearly that this defect is NOT a cavity, NOT a crack but an infection of a fungus. The size and position of the infected area can be estimated by drawing a line around the green dots. In the left tomogram nearly the whole tree is infected.

In contrast to tomograms of cavities the sonic velocity does not slow down continuously towards the centre of the trunk and raises again towards the edges.

7.9.2. Coloured or grey tomogram

Very early versions of the PiCUS program (1997 – 1998) used a grey scale to draw the tomogram.

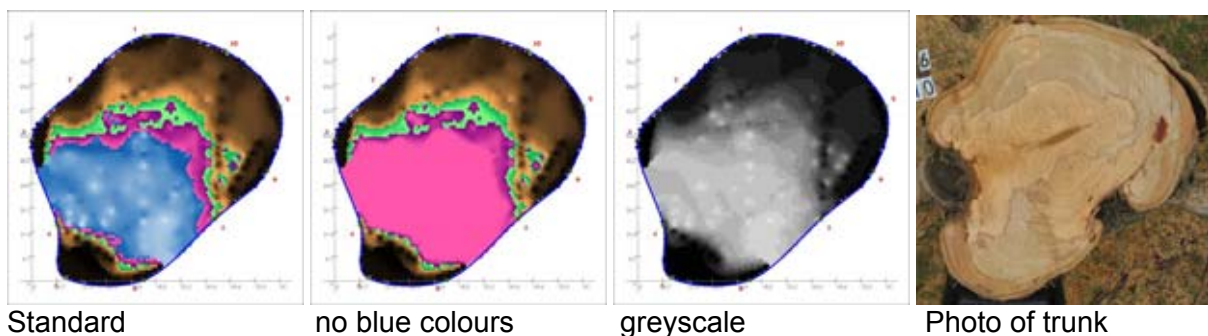
Because it very difficult to determine the exact extend of the damaged area in a grey tomogram we are now using colours to display the areas of different density.

The use of “sharp” colour borders is difficult because the sonic velocities vary by different species, seasons, water contents etc, sometimes even within the same tree.

In order to avoid this problem we are now working with “relative velocities”. This means we compare all velocities recorded in one level with each other. Doing so, we can show where the sonic waves travel slower compared to the “fastest” areas.

As mentioned above there are three main colour groups to pay attention at when analysing the tomogram. The violet / blue / while colour group can be shown in violet colours only. This may be useful for easier interpretation.

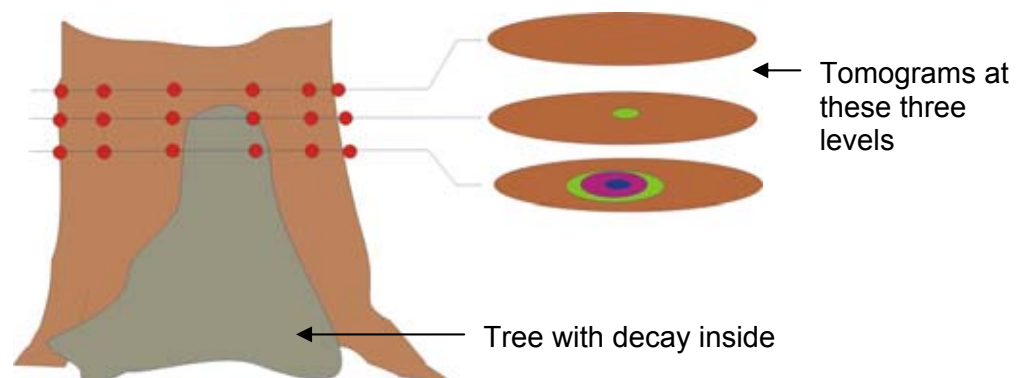
The examples show a tomogram in standard, violet only and grey colours.



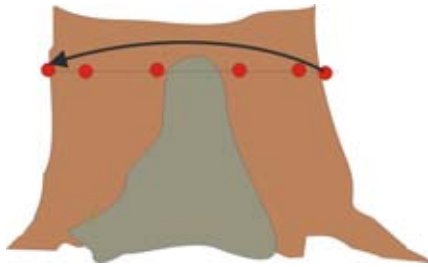
7.10. Vertical extension of damaged areas

The tomograms recorded show damaged wood in the level of sensors only! All other damaged wood above or below the level of tomography are not taken into consideration.

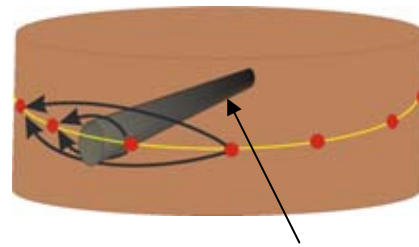
The following sketch shows the influence of sound wood in the tomogram at the vertical edge of decay.



Please note that the tomogram in the middle shows less decay than present in the tree. This is because sound waves travel in all 3 dimensions through the tree. Therefore the sound wood over (or under) the end of a decay “hides” the decay in the tomogram near the end of the decay. The following sketches show the way the sound waves travel:



Sound waves at vertical end of decay



Sound waves passing a small vertical hole

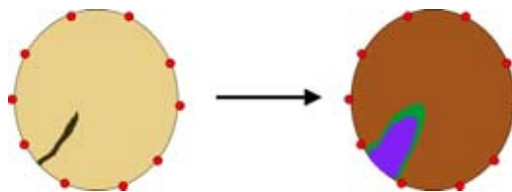
7.11. Special cases

If you interpret a tomogram please mind the following special cases:

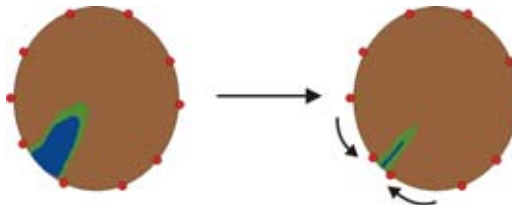
1. Wetwood - (Populus vs. Aesculus). If a tree suffers from wet wood, the tomogram sometimes shows the wet wood area like being hollow.
2. Cracks are shown wider than they are. This is because the resolution at the edges (!) of the tree slice in the tomogram is limited to the distance between sensors. If necessary, make smaller the distance between the adjacent sensors and repeat the sonic measurement.

Tree with crack

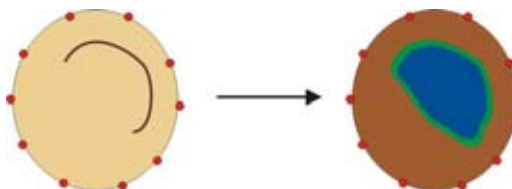
Tomogram of this tree



If necessary, this result can be improved by moving the sensors next to the crack closer to the crack (like shown in the sketch) and repeating the measurement:



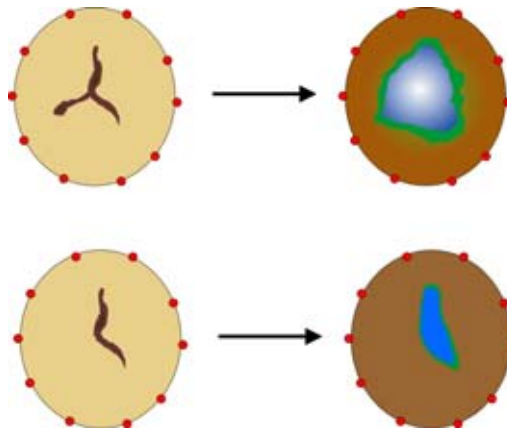
3. Ring shaped cracks - Eucalyptus trees tend to have ring shaped cracks.



4. Star shaped cracks - because the sonic waves "circle" the crack. Old oaks tend to have cracks like this.

Slice of a tree with "Star shaped" crack

Tomogram of this slice



The following example shows a “star shaped” crack caused by included bark. It is a Red Oak (North America), consisting of three trunks.



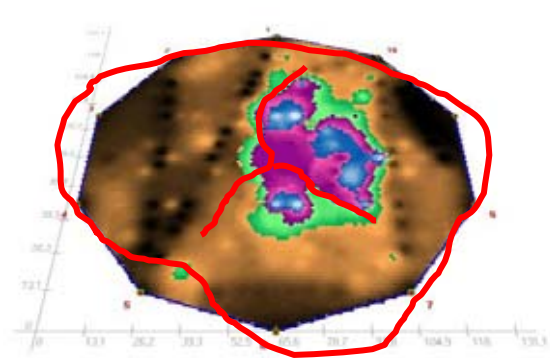
Three Trunks of a Red Oak



Stump of this tree

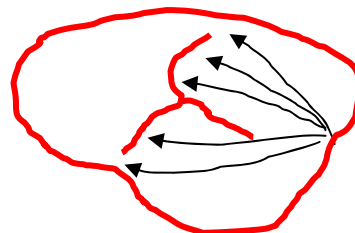
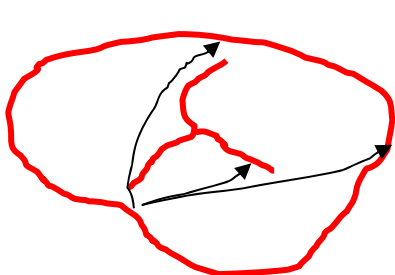


Stump with highlighted included bark

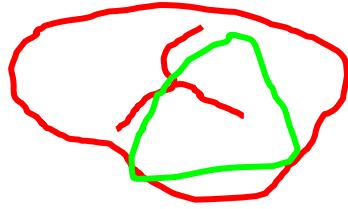


Tomogram of this tree

Sonic waves have to „circle“ the included bark:



During a measurement the sensors record the time the sound waves need to travel from tapping point to the other sensors. Included bark does not transmit sound waves very well, if the bark is thicker, it does not transmit the wave at all. Therefore, the sound waves have to circle the bark and need a longer time. The PiCUS can detect convex structures only. The best approach in this situation is that:



7.12. Meaning of the legend

By pushing the “Show Legend” button after calculation B a value indicating the ratio of damaged to sound area of the tree is shown. You have to configure the legend in configuration menu first before you can use it.

The total amount of pixels is counted during drawing the tomogram. Furthermore, the number of brown pixels is counted. This number in relation to the total number of pixels is the value shown as “sound wood”. The number of violet / blue / white pixels in relation to the total number represents the amount of damaged wood.

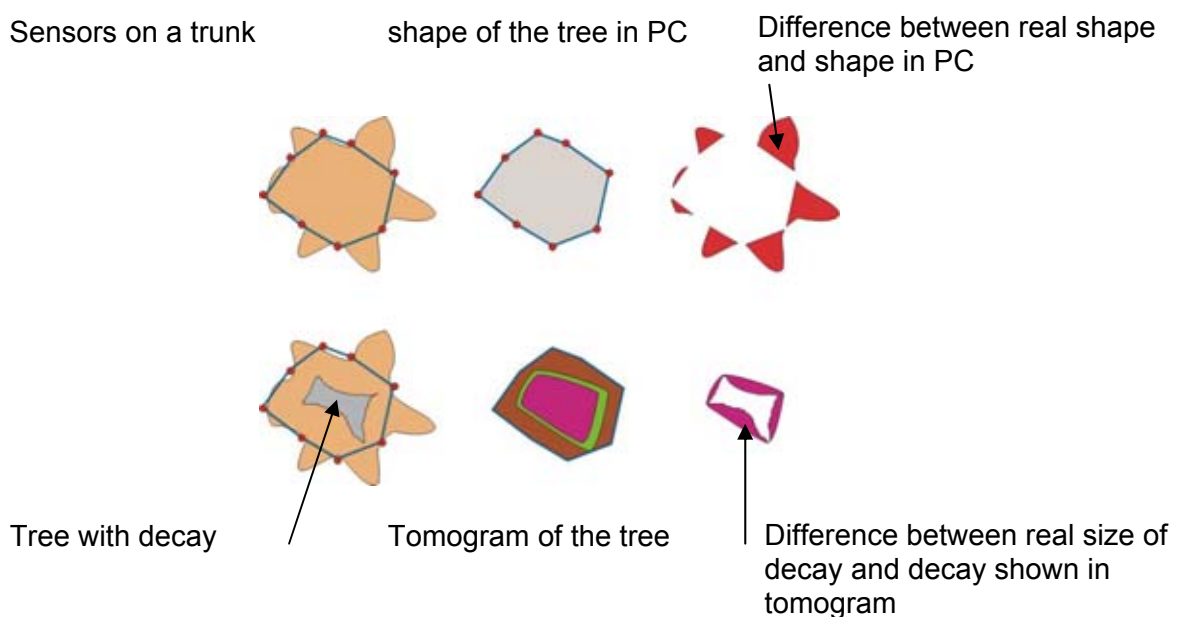
The number of green pixels is not counted.

The values for sound and damaged wood are rough numbers only! The accuracy of these values depends on

1. Shape of the tomography level of the tree
2. Position of sensors
3. Shape of the damaged area
4. Reason for the damage (Crack, decay, cavity)

The following sketches show the difference between the real size of the tomography level and the total number of points counted. That is why the total number of pixels is smaller than the real size of the slice of the tree.

The computer does not have correct information about the real shape. Therefore, the value of sound wood given in the legend is usually too small.



There might be a difference real size of the damaged area and the size shown in tomogram depending on the shape of the decay and the measuring principle of the PiCUS. In particular we have this case when the damage shown in the tomogram is caused by cracks.

7.13. Questions about tomograms? Send data to our service!

If there are any questions that cannot be answered by reading the manual, please do not hesitate to contact our service team.

We are always interested in results of the PiCUS measurements of our customers!

To be able to service the question, please provide a complete set of the following information:

1. Take digital photo of the tree. If possible take a photo from the distance and from a closer position showing the measuring level and the fungus or cavity etc.
2. If the tree has been felled, please take a photo of the stump (taken exactly vertically from above) in the measuring level also. Mark the position of the sensors on the stump before taking the photo.
3. Send all photos and the PiCUS data file ("name_of_the_tree.pit") to our email address: picus@argus-electronic.de. Please use convenient names for the files:

For instance:

Oak nr 123.pit	- PiCUS data file
Oak nr 123 total view.jpg	- photo of the standing tree
Oak nr 123 stump.jpg	- Photo of the stump

4. Write a short description containing tree information like
 - Species of the tree
 - Date of the measuring
 - Reason for the measurement (visible damage / decay / fungus etc)
 - Weather conditions (spring, summer, temperature of the last days, moisture contents of the soil...)

8. PiCUS Rootfinder

8.1. Introduction

The PiCUS hardware is capable to help in other measuring situations. One of it is getting information about the roots of a tree.

In order to get information about the roots of a tree the PiCUS sensors need to be placed on the trunk of the tree near ground level.

Sonic impulses are generated by using a metal plate and/or metal pole and a hammer on known positions around the tree. The time of flight of the sonic impulses is measured by the PiCUS. By analysing the times of flight this measurement gives limited information about the existence of large roots. This works in a few situations only.

The method is working best on solitary trees, without over or beneath ground level buildings, pipes etc. .

The investigations are strongly influenced by soils of different compression, different water contents, large objects in the ground (stones etc). Because of all these influences the use of these measurements is often limited.

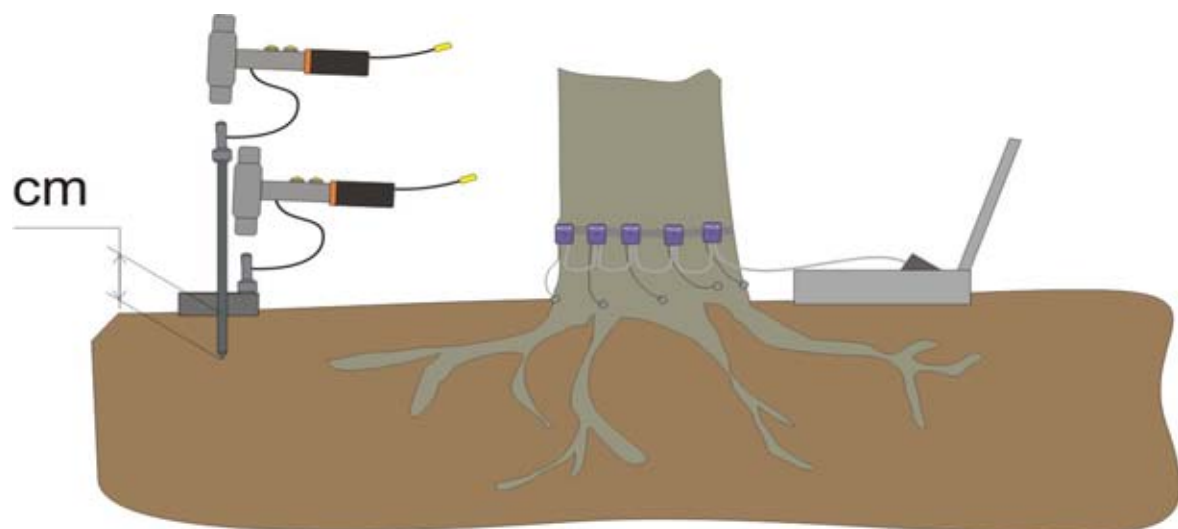
Compared to sonic measurements on trees it is difficult to generate and measure sonic waves in the soil – tree combination. This is because the sonic waves have to overcome many material borders from sender to receiver:

Hammer → Metal plate or pole → Soil → Root → Nail → Sensor.

Every time the signal has to leave one material and enter another material amplitude and frequency of the signal are changed. Measurements like this do not allow to give information about the health condition of roots. In a few cases, we can obtain limited information about the soil or about the connection between roots and soil.

8.2. How to find large roots

1. Mount sensors closed to ground level on the tree (like doing sonic measurements of the trunk). The last module of the chain is used for transmitting sound waves (starter). Connect this last module to the chain by using the long PiCUS data cable. In order to connect the chain to the PiCUS power unit the cable of the first module is used. You should use a hammer and a metal plate (diameter $\geq 10\text{cm}$, thickness $\geq 1\text{cm}$) to generate sound signals. In some situations it is useful to use a metal pole to generate sonic signals below ground level. The measuring site should look like this:



2. Open a new „Root“ – File:

„File“ → „New Root file“

3. Enter positions of the sensors on the trunk (similar to PiCUS Sonic Tomograph measurements).

„Measurement“ → „Tree Geometry“ → „Free shapes Calliper“

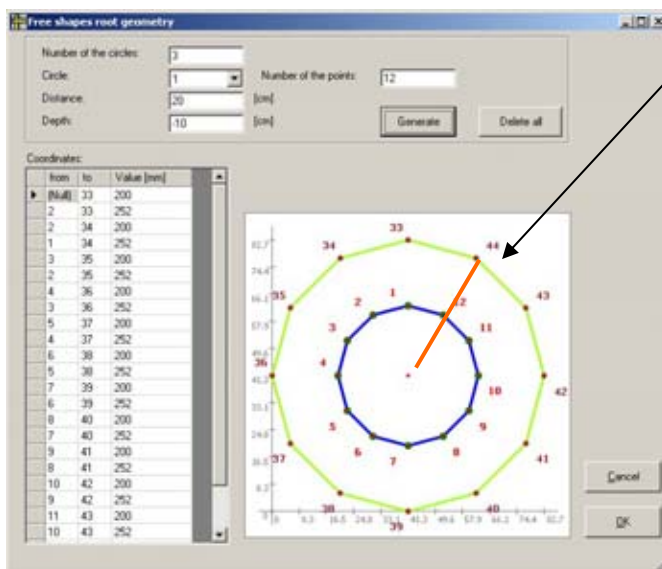
When entering the number of modules use the number of sensors on the trunk. The additional module that is used for signal generation is not considered at this point.
Example: If you place 8 sensors on the trunk, you have to enter „8“ in the geometry window. The module number 9 is used for signal generation only.

4. Enter the positions of the points on the ground where you want to generate sonic signals. Use the Following window:

„Measurement“ → „Root Point Geometry“ → „Free Shapes“

All points on the ground to hit on are spaced out evenly on circles. You can enter a depth for each circle. If you enter a depth of some centimetre, you have to use the metal pole (watch sketch) for tapping. Enter the distance of the circles from the tree and set the number of measuring points. Click on „Generate“ to create the „circle“ of measuring points (root-points). The positions of all „root-points“ are now shown in the table.

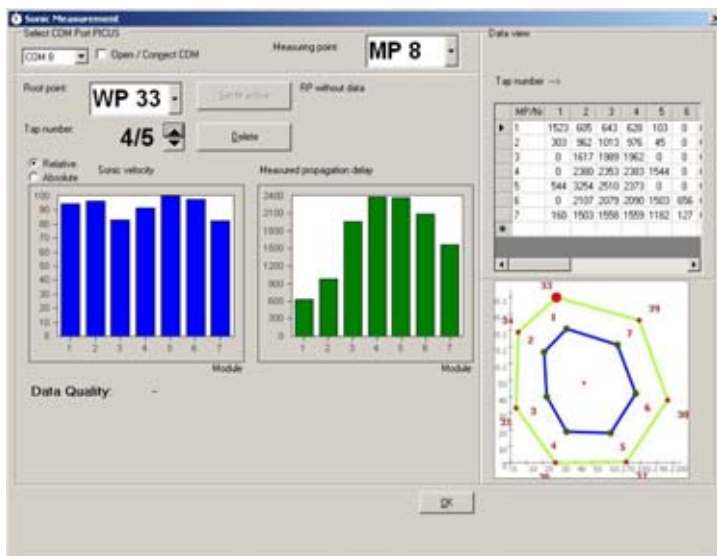
„Root-points“, MP on the trunk and centre of the tree lay on a line if the number of MP is equal to the number of „root-points“ on the ground.



6. Set Nr# of circles and „root-points“
7. Select current circle
8. Set distance to the tree
9. Enter depth of tap
10. „Generate“
11. Repeat 2-5 for all circles

5. Start sonic measurement

„Measurement“ → „Measure Sonic“



1. Open COM port
2. Select module that is used for measuring only. In this case it is 8.
3. Select root-point
4. Hit on metal plate / pole
5. Repeat on all root points and circles

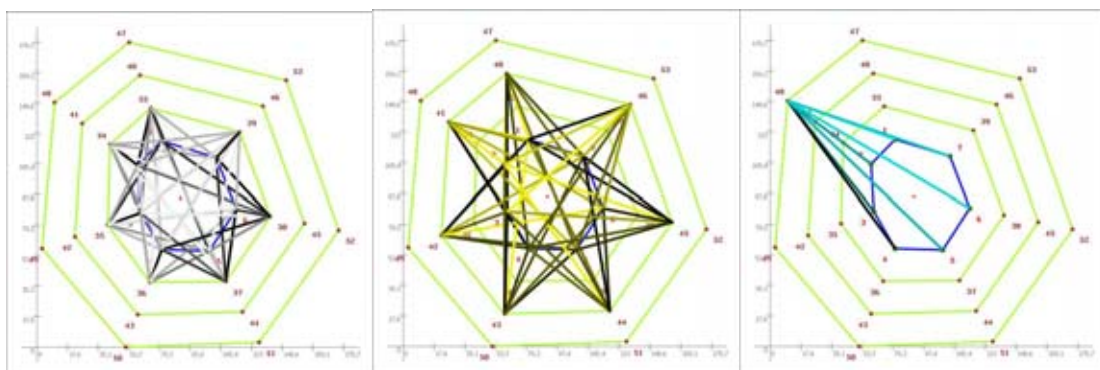
6. Select the module that is needed for tapping to give the starting signal. This module is uses on all root-points. Open serial port and turn on the PiCUS hardware.
7. Select the root-point you intend to start with. In our example it is "WP 33". Put the metal plate on the correct position. Hit on the plate. Use a larger hammer instead of the PiCUS hammer. If you want to create sonic waves underneath the surface you can use a metal pole (watch sketch). Despite the very sensitive PiCUS sensors you may need to hit relatively hard on the plate.

8.3. Presenting of the results

The presentation of the results is similar to Calculation C when doing tree trunk investigations. Click on this Icon / Symbol to start calculation:



The velocities measured are shown for each circle using another colour – or strictly speaking – a colour – black combination. The darker the colour is the better / faster the sonic signal was travelling between the points.



The example shows a tree that was investigated in three circles.

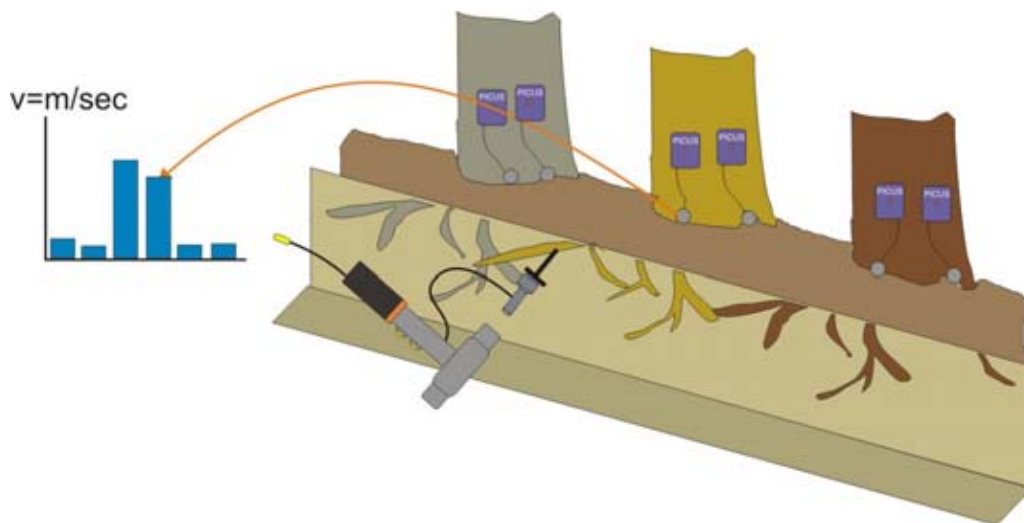
8.4. Root – Tree allocation

A tree has been called into question for a building subsidence claim. A test pit has been dug at the perceived point of interaction with tree roots to the building. This will be close to the building. Roots have been uncovered and now they need to be identified and allocated to one of the nearby trees.

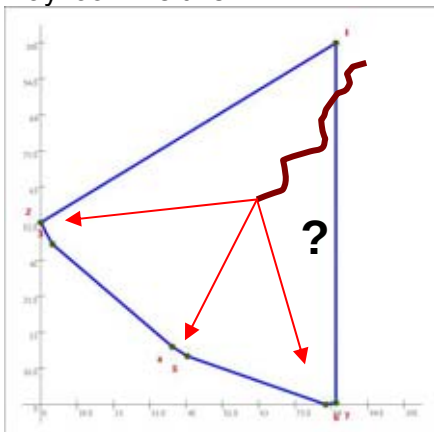
This question may be answered by the help of a measuring using the PiCUS hardware in some cases. There is a minimum size of the root that limits the application. Also, the distance to the tree must not be too large. Both values do depend on many influences. It is difficult to specify the limits. A raw estimation is that the root should be of 4-6 cm in diameter minimum for this sort of measurement. The longer the distance to the tree is the larger the root needs to be.

Please proceed as follows:

1. Mount at least one sensor (better 2) on each tree the root may belong to. Apply sensors near ground level as shown in the sketch.
- 2.



3. Use another sensor / module as a starting signal. Tap a nail into the root.
4. Create a new file for a "Tree trunk tomography" investigation despite you intend to work on a root: „File“ → „New Tree Tomogram“. Disable "Smooth Tree shape" in configuration menu tabulator "calculation 1"
5. Enter the positions of all sensors by using the „Free shapes with Calliper“ method. Alter the base-points suggested. If sensor 1 is used as tapping point (on the root) you should select nr 1 and the last sensor to be used as base-points. Use the measuring tape to measure all distances. Enter cm instead of mm! The measuring site in the "*.pit" data file may look like this:



Example of an investigation with 3 trees.

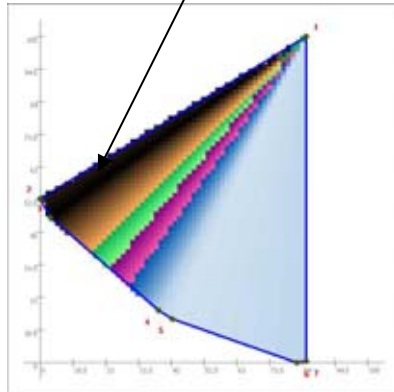
MP 2,3: mounted in tree nr 1,
MP 4,5: mounted on tree nr 2,
MP 6,7: mounted in tree nr 3.

Base points for geometry measurement are 1 and 7.

Module 1 is used for start signal generation on the root.

1. Carry out the sonic measurement. Tap on position nr 1 only. In most cases, only the tree the root belongs to will receive sonic signals. All other tree will receive very slow signals or no signal at all.

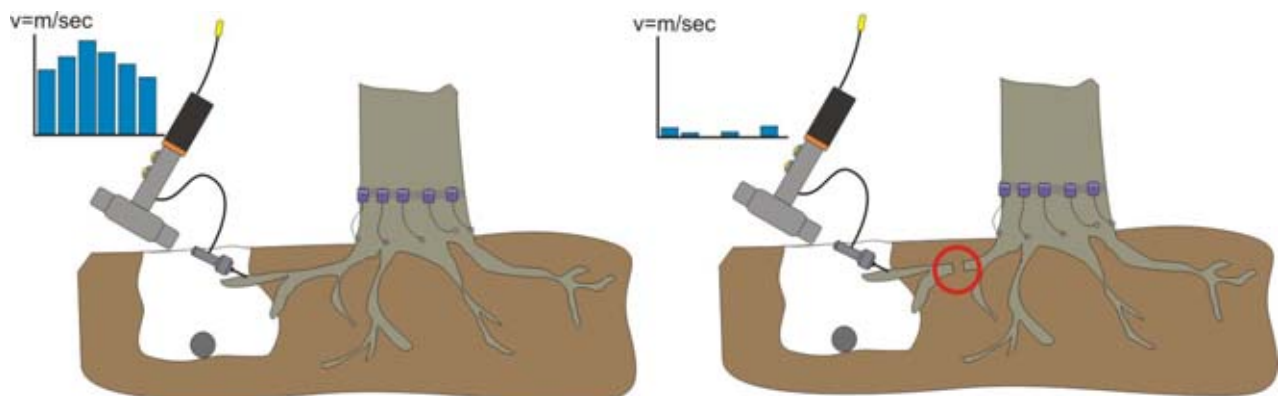
2. Disable "Data Checking" in configuration menu before the calculation is called. Run "Calculation C" to view the result.
3. In this example tree nr 1 receives sonic signals best. The root is probably connected to tree nr 1.



Please mind that the roots of tree can be in good contact to each other. The result may be less unambiguous in these cases.

8.5. Root cut off?

In some cases the PiCUS hardware can be used to find out *if* a root is connected to a tree. The following sketch shows the measuring site.

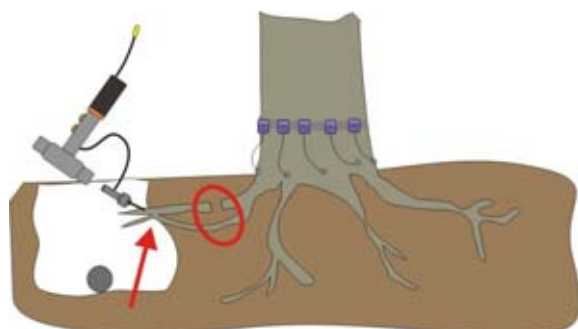


Proceed the sonic measurement like described in the root mapping chapter. You need to tap on one "root-point" only.

Tap directly on the pin in the root and on a metal plate near by the root. There should be a significant difference between taping on the root and on the soil.

If there is NO difference the root does not transmit sonic waves very well. This is an indication for a defect.

Please mind that roots do not follow known rules when growing. They can split off and grow together again. In these cases the result of a measurement is less accurate.

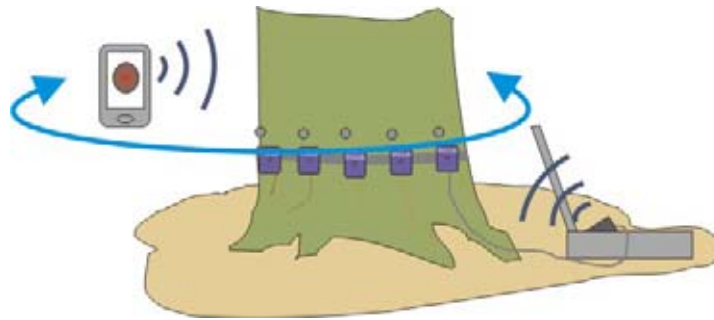


9. Technical Information and Trouble shooting

9.1. PiCUS® with Bluetooth™ - wireless radio technology

9.1.1. General information

In order to improve the handling of the Pocket PC during sonic measurements PiCUS® Sonic Tomograph is available with integrated Bluetooth™ module. Using this feature the tree expert can move freely around the tree during a measurement with the Pocket PC in his hand without being hindered by a cable.



9.1.2. How to use the Bluetooth™ connection

Before you can use the Bluetooth™ connection between PocketPC (or PC) and the PiCUS® both PiCUS Bluetooth™ module and PocketPC/PC Bluetooth™ module must be initialised.

At delivery time, of PiCUS® the PiCUS Bluetooth™ module is initialized and ready for work.

In order to activate the Bluetooth module in the power supply the USB cable need to be removed BEFORE the power supply is tuned on.

The blue LED starts to indicate after turning on the PiCUS® power supply. After 3 seconds the Bluetooth™ connection is ready for work.

To establish the Bluetooth™ connection the Pocket PC or PC has to be set up also. The following steps depend on the model you are working with. Please go to the proper section of the manual.

9.1.3. Setup of hp iPAQ PocketPC h4100 for Bluetooth™ connection to PiCUS

The following steps are necessary to establish a Bluetooth™ connection between the PiCUS and hp iPAQ PocketPC h4100:

1. Turn OFF PiCUS Sonic Tomograph.
2. Disconnect the USB cable from the power supply
3. Turn on HP4100. Activate Bluetooth functionality of the PocketPC by tapping on the Bluetooth symbol (at the bottom, right side) and choose „Bluetooth ON“
4. Tap on the Bluetooth symbol again and choose „Bluetooth settings“. On the following cards please you can configure the functionality of the Bluetooth system of the PocketPC. You do not need to repeat this procedure at each Bluetooth start.
5. Card „General“: no changes.
6. Card „Accessibility“: Set checks at “Allow other devices to connect” - “All devices” and “Other devices can detect me”.
7. Card „File transfer“: no needed. Turn this service off if you do not need it.
8. Card „Information exchange“: no needed. Turn this service off if you do not need it.

9. Card „Serial connection“: Set check at „activate“. Do not choose „Authorisation required“! Do **not** choose „Authentication required“. Keep in mind the COM ports: Incoming - COM 8, Outgoing - COM 6
10. Card „Network“: no needed. Turn this service off if you do not need it.
11. Card „PAN Server“: no needed. Turn this service off if you do not need it.
12. Click “OK” on top of window to close this setup.
13. Turn PiCUS Sonic Tomograph on. Bluetooth LED indicates. After 3 seconds, it is ready to work.
14. On the PocketPC tap again on the Bluetooth symbol and start the Bluetoothmanager.
15. Tap on „new“ to show Bluetooth devices around you. The Connection assistant is being started. Search for available services of the „Bluetooth devices“ (first item) and tap “next”. In order to look for other devices again please tap on the small symbol with the two green arrows.
16. If PiCUS is turned on the symbol of „Picus Sonic Tomograph“ is shown after a few seconds on the PocketPC. Tap on this symbol to check the available services of this device.
17. After finding the service „Picus serial Port“, please select it and tap „next“
18. The shortcut is now ready.
19. In the PiCUS program of the PocketPC's you can now select the Bluetooth - COM port in the menu COM - configuration. Choose COM port 6. After setting the check next to the drop down menu the COM port is being opened.
20. The Bluetooth manager of the PocketPC is shown automatically now. Choose the connection to the PiCUS by double tapping the PiCUS shortcut icon. The blue LED of the PiCUS power supply shows light without indicating. Now you can select the number of modules connected and click “Test” to test the connection.

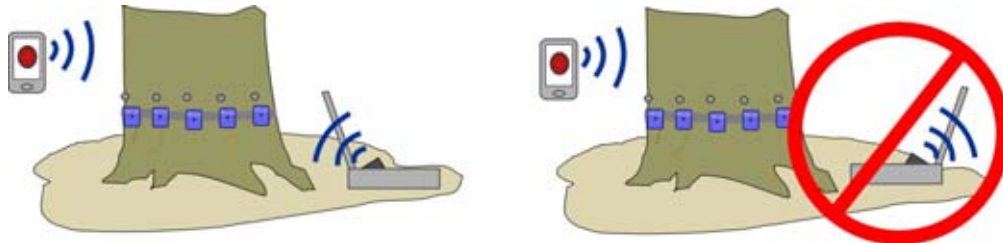
The connection is now ready to work.

NOTE: PLEASE do not turn off either PiCUS or POCKET PC while a Bluetooth connection is active. Please terminate the connection first.

If the Pocket PC does not respond as described above, try to reset the Pocket PC by pressing the small reset button. Please mind: All unsaved data will be lost.

9.1.4. Radio shadow

To avoid a “radio shadow” on very huge trees please twist the PiCUS suitcase on the ground as shown in the sketch.



9.1.5. Setup PiCUS® Bluetooth™ Module

At the delivery of PiCUS the PiCUS Bluetooth™ Module is initialized and ready for work.

If PiCUS® is turned off during the boot process of the Bluetooth™ Module the setup information of the Bluetooth® Module can be deleted in rare cases. Therefore, PiCUS® must operate for 3 seconds or more before it can be turned off.

If you do not wait for 3 seconds, the setup information of the Bluetooth® Module could be deleted.

In that case the PiCUS® Bluetooth™ Module has to be reinitialized.

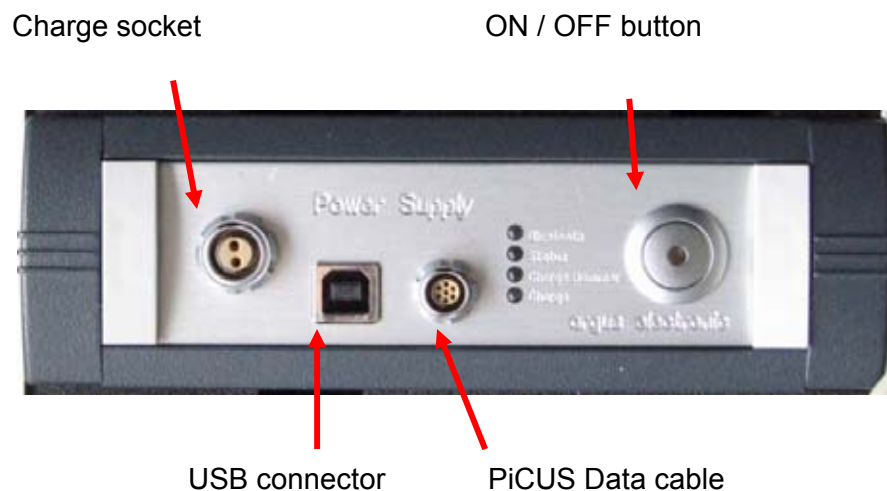
To initialize the PiCUS® Bluetooth™ Module please proceed as follows:

1. Turn off the PiCUS®.
2. Connect the PiCUS® power supply to the PC the PiCUS® analyse software is running on. Use the provided serial cable for this connection. You can use the Pocket PC also.
3. Put the radio switch on the PiCUS® power supply in upper position (Bluetooth™ radio ON)
4. Start the PiCUS® PC program and select the COM port the PiCUS® is connected to. To select the COM port please Click to “Configuration” in the main menu and than choose “Select Com port”. Make sure that no other application on the PC uses the COM port you want to work with. Close the COM port select window after having selected the COM port.
5. Open the “Bluetooth Setup” window by clicking on “Configuration” (Main menu) and than “Bluetooth Setup”.
6. Press the “Start Bluetooth Setup” button.
7. The program ask you to turn on PiCUS. Turn ON PiCUS®. After approx. 2 seconds a message is displayed: „+++ Press <CR> <CR> ... “. The setup continues automatically. Please do not press any key until the program asks you to do this.
8. If the setup is complete a message box appears. After turning off PiCUS the setup is ready. The PiCUS® Bluetooth™ Module is ready to work now.

9.2. PiCUS Power supply

The PiCUS power supply (manufactured after May 2007) offers USB connection to the PC and charging functions for the Electronic radio hammer (PRH).

- Auto power off function: after 10 to 15 minutes of idle operation the power supply turns off itself.
- Low voltage power OFF: The power supply turns off automatically when the battery power is too low. A warning message is shown on PC – PiCUS program window a couple of minutes before low-voltage-power-off is performed.



LED description (Top first)

- Blue LED to indicate Bluetooth status
- Yellow Status LED
- Green LED to indicate Hammer (PRH) charging
- Green LED to indicate PiCUS charging

9.3. Accumulators and Charging

9.3.1. Charging the PiCUS

To charge the PiCUS please proceed as follows:

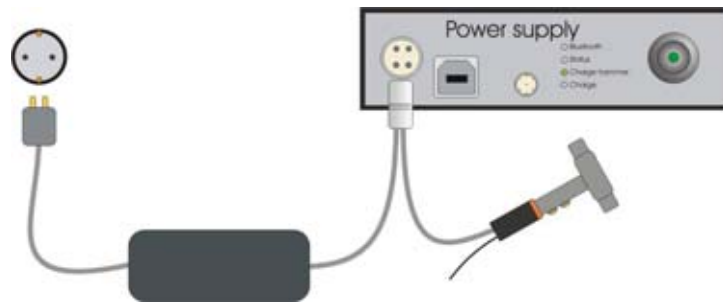
Do NOT operate the PiCUS without batteries connected. The charger MUST not be connected to the PiCUS power supply if batteries are not connected.

1. Connect the AC/DC converter supplied to a power socket.
2. Connect the DC plug to the PiCUS charging socket.
3. Charge LED shows green light when charging. Accumulators can be charged while measuring.
4. The accumulators are loaded when the charge LED is off. Depending on accumulator status, this may take up to 8 hours.

9.3.2. Charging the electronic radio hammer (PRH)

The PiCUS radio hammer (PRH) needs to be charged. When fully charged it will continually work for at least 10 hours (at 20 °C). To charge the hammer please proceed as follows:

1. Connect the AC/DC converter to the PiCUS power supply (large connector).
2. Connect the AC/DC converter to the power outlet (110/230V AC, 50/50Hz)
3. Connect the PRH to the small connector of the AC/DC converter cable.
4. The “charge hammer” LED shows green light.
5. Charging takes up to 4 hours the LED is turned off.



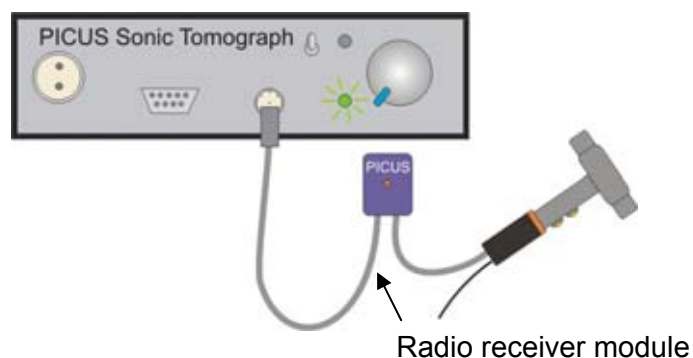
9.3.3. Charging the electronic hammer (PRH) with old PiCUS Power Supply

The PiCUS radio hammer (PRH) needs to be charged. When fully charged it will continually work for at least 10 hours (at 20 °C).

If the PRH is operated with an old PiCUS power supply (build before 7/2007, serial communication) the charging procedure for the PRH is as follows:

1. Connect the PRH to the small connector of the radio receiver module.
2. Connect the longer cable of the radio receiver module to the PiCUS Power Supply
3. Turn ON the PiCUS power supply to start charging. The “charge hammer” LED INSIDE the radio receiver module shows light.
4. After up to 4 hours the LED is turned off. Hammer charging is done.

The batteries of the PiCUS are discharged by approximately 10% when the PRH needed a full recharge.



9.3.4. Changing accumulator

If the accumulator pack needs to be replaced please proceed as follows:

1. Turn off PiCUS and disconnect all modules and cables from the power supply.
2. Remove the soft cover.
3. Disconnect cable from old accumulator.
4. Connect new accumulator.

9.4. Operation of the PiCUS Radio Hammer (PRH)

9.4.1. Radio transmission problems

The radio transmission between the radio hammer (PRH) and the radio receiver module (that is either inside the PiCUS power supply or in an extra module) works over a distance of many meters. However, there might be transmission problems if there is a big tree right between the PRH and the receiver module.

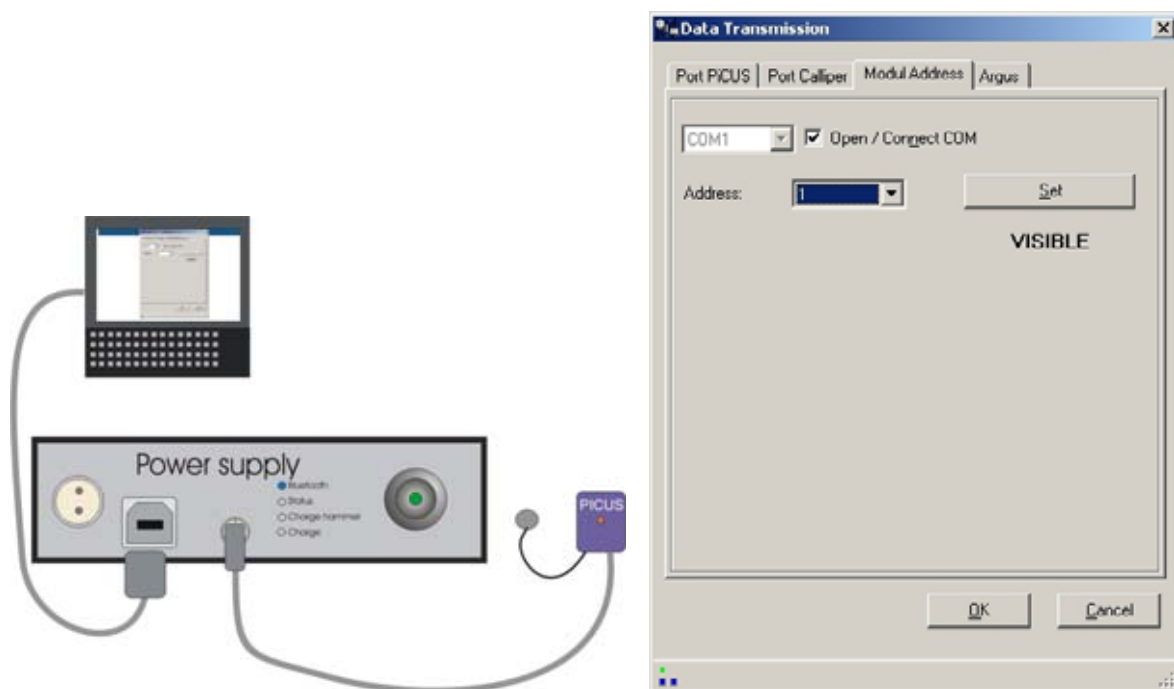
The hammer - transmission - status LED of the power supply will start to flash to indicate the loss of the connection.

In such a case, we recommend to move the receiver module (the PiCUS case, where the power supply is installed in) around the case to get in “eye contact” to the hammer.

9.5. How to change address of modules (manufactured after 8/2007)

The address of PiCUS modules, manufactured after 8/2007, can be changed by software. Please proceed as follows.

1. Turn OFF the PiCUS.
2. Connect the module to the PiCUS power supply. ONLY ONE module must be connected at the same time!
3. Connect PC and PiCUS power supply by serial, USB or Bluetooth connection.
4. Start PiCUS PC program version Q71 (7/2007, or higher version).
5. Open the module configuration window:
“Configuration” → “Select COM Port” → Tabulator: “Module address”
6. Turn ON the PiCUS.
7. Open the COM port in the “Module address” window.
8. Select the address (1 through 30) that you want to program from the drop-down list.
9. “Left-click” on the word “visible”. The button “set” is shown.
10. Push the “set” button to program the new address into the module connected.
11. The programmed address is shown.
12. Remove the module from the power supply. Connect the next module to the PiCUS power supply and proceed similar to change the address.



9.6. No connection to Modules

9.6.1. Cabling

If the software does not find the modules the cabling has to be checked first.



- Connect PiCUS Power supply with the serial port of the PC or PocketPC by the serial cable provided
- To exclude errors in the long data cable connect the module – chain directly to the PiCUS power supply
- Connect the modules to each other

9.6.2. Electrical test

After all cabling is done turn on the PiCUS® system by twisting the power button anti-clockwise:



The red marked LED must turn green.

Immediately after switching on the modules start flash – number 1 first, followed by #2 , #3 and so on. If this test succeeded, the basic electrical connections are made.

To check the modules please proceed as follows:

1. Start the PiCUS.exe
2. Open the COM Settings Window
3. Select number of modules connected
4. Select the COM Port
5. Open COM Port by setting the check at “COM open / connect”
6. Push „Test“ to test the connection. The screen should report „Modules o.k. : 1,2,..10“.
7. If not, turn the system off, remove the module number 1 from the chain and connect the 2ed module directly to the power supply. Turn on power, wait for all modules doing their initialisation and press „Test“ again.

8. If there is still no connection, there is probably a problem with the serial cable or the COM port. Try to use another cable first, if there are still problems try another PC or the PocketPC.
9. Call *argus electronic* service hotline if you can not solve the problem

9.6.3. Software / Active sync

Before starting PiCUS.exe make sure of no other program uses the COM – port that you intend to use for the PiCUS program.

If you have installed the „Active Sync“ program, make sure that Active Sync is using another port. If you got only one COM port (for instance at notebooks) disable the „Active sync“ program by the following commands

1. Open the Microsoft Active Sync Window by pushing the small green or grey circle –



symbol with arrows.

2. Turn to File → Connection settings
3. Disable the connection via the port you intend to use for the PiCUS program. The icon

turn to: 

9.7. Maintenance

The PiCUS Sonic Tomograph is to be maintained by *argus electronics gmbh* Rostock, Germany. The device does not contain any parts that can be maintained by the user. If the device or parts of it should be opened by a person not authorized by the manufacturer, all warranty claims are void.

9.8. Shipping of PiCUS® Sonic Tomograph

If you want to send the PiCUS via parcel services you **MUST include** lots of crumpled newspapers (or similar materials) **IN THE PiCUS** suit case!!! This is to avoid damages caused by vibrations during transport. The photo shows an example.



Please do not send nail boxes. If you want to send nail boxes, please close the boxes with extra adhesive tape!

9.9. Pocket PC hints

9.9.1. Compatible Pocket PC

1. HP iPaq h4100
2. HP iPaq 5550
3. HP iPaq Hx4700
4. HP iPaq Hx2410
5. HP iPaq Hx2750

6. Dell Axim X50 (Bluetooth inside)
7. SPV M1000 smart phone running Win2003CE with a PXA263 processor.
8. Orange SPV M2000

9.10. PC System Requirements

The PiCUS software requires the following PC hard and software (minimal):

PC Hardware:	PC Pentium 2 or higher, 450 MHz or higher
PC Operating System:	Microsoft Windows 2000/XP (all updates loaded), Vista
PC Screen resolution:	1024 x 768 (or higher)

PocketPC operation systems: Windows mobile 2003, 2005

Please choose 65000 colours or more on your computer to view the PiCUS tomograms in full resolution.

9.11. Technical data of the PiCUS radio hammer PRH

Type of Accumulator:	NiMH
Charging time:	4 hours
Time of operation:	10+ hours (depending on temperature)

9.12. Technical data

Dimensions of the unit:	540 x 480 x 180 (B x T x H) [mm]
Weight of the unit:	11.3 kg

Power supply:	Charging unit at 100 – 240 V~ (AC), 50 Hz
Type of Accumulator:	NiMH
Accu power:	The accumulators get older. Therefore, the age and number of charging/discharging procedures reduces the capacity.
Charging time:	max. 4 hours, (no overloading because of automatic turn off)
Power consumption of one PiCUS module:	approx. 35 mA

Time of operation at 20°C:	
with 8 Modules:	approx 7,5 h, meets 80 trees: 5 min / tree
with 10 Modules:	approx: 6 h, meets 50 trees: 7 min / tree
with 12 Modules:	approx: 4 h, meets 24 trees: 10 min / tree

Operation temperature:	0°C to 40°C
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RS232 settings:	9600/8/N/1 no hardware control
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Humidity:	Electronic devices can be damaged by humidity. The PiCUS Sonic Tomograph should be protected against the direct impact of water and humidity. Nevertheless, the system is protected against light rain.
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Release date of this manual:	09 October 2008
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Disclaimer: Technical specifications are subject to change without notice.

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Prof. Dr. Steffen Rust
Dr. Bernd Gustke, Institut für Gehölze & Landschaft GmbH

10. Address

The PiCUS Sonic Tomograph was developed by two German companies:

argus electronic gmbh and ***Institut für Gehölze & Landschaft Dr. Gustke GmbH***.

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